HyperQUEEN: Hyperspectral Quantum Deep Network For Image Restoration

Chia-Hsiang Lin, *Member, IEEE*, and You-Yao Chen, *Student Member, IEEE*

***Abstract*— Quantum science just winning the 2022 Nobel Prize in Physics must lead future development of remote sensing technologies. However, given the very limited number of entangled quantum bits (qubits) even in the most advanced quantum computers, is processing remotely sensed hyperspectral images (featured by its large data volume) using quantum computers technically feasible? Even if the quantum image state can be well processed to the quantum state of the target image (QSTI), it cannot be perfectly retrieved/output as the QSTI will collapse to some eigenstate once it is measured. Owing to these challenges, current quantum image processing technologies can only achieve classification-level applications requiring just a few output qubits. We design a hyperspectral quantum deep network (HyperQUEEN) to encode the hyperspectral information using very few qubits, as well as to learn the mapping from some measuring statistics (associated with the collapsed-QSTI) to the target image (instead of directly retrieving the unobservable QSTI), thereby solving the challenges. HyperQUEEN is the first quantum architecture that makes a breakthrough to blindly reconstruct the National Aeronautics and Space Administration's (NASA), Washington, DC, USA, damaged hyperspectral images, which means a lot for the upcoming space era. As the immature quantum facility nowadays does not yet allow us to fully exhibit its high potential, we are not aiming at developing state-of-the-art methods, but are demonstrating the feasibility of quantum hyperspectral remote sensing. Mathematical analysis guiding our design toward the low-rank quantum deep network, together with comprehensive experiments, will also be reported.**

***Index Terms*— Blind signal processing, hyperspectral remote sensing (HRS), quantum computing, quantum deep learning, quantum image processing.**

I. Introduction

HYPERSPECTRAL remote sensing (HRS) plays a critical role in the upcoming space exploration era and is greatly supported by all kinds of hyperspectral image processing technologies, including hyperspectral source detection, hyperspectral super-resolution, hyperspectral compressed sensing, hyperspectral dehazing, and hyperspectral target detection [1], [2], [3], [4], [5], [6]. Remotely sensed hyperspectral images are usually with wavelengths widely ranging from 0.4 to 2.5 μm having approximately 10-nm spectral resolution. Such densely populated spectral bands result in much larger data volume when compared with typical image types, but the induced hyperspectral information is critical to uniquely depicting and discriminating the material identity. Remotely acquired hyperspectral images are often with lower pixel resolution, so one pixel would contain multiple materials. Thus, the hyperspectral information of different materials may be mixed together in one-single pixel. Recent advances in the hyperspectral unmixing theory and algorithms developed in the mathematics and machine learning societies can be adopted to recover the pure hyperspectral information for every single material (often unique) [7], [8], [9], thereby facilitating the subsequent applications of material identification and classification in hyperspectral space exploration.

The Royal Swedish Academy of Sciences, Stockholm, Sweden, has just awarded the 2022 Nobel Prize in Physics to scientists pioneering quantum information science, which has great potential to lead future developments of various remote sensing technologies. To get a sense, a specific sampling task can be completed in about 1.2 h by the Zuchongzhi quantum computer, while it would take about eight years in classical computers [10]. In fact, quantum computing has numerous applications in solving NP-hard problems, such as the automatic detection of financial crime from the transaction records [11]. Quantum computing is very good at reducing the computational complexities of typical combinatorial problems, such as implementing fast search via Grover's algorithm [12], or finding the prime factors of an integer in the cryptography via Shor's algorithm [13]. In space exploration, the outstanding capability of hyperspectral imagery in material identification and quantification may be combined with the analysis of the hardly tractable interaction among molecules, for which the required huge computational loading can also be realized by the quantum computing [14]. Even with the above-mentioned theoretical evidences, we should remark on the practical aspect that unless there is a theoretical breakthrough in realizing quantum error correction (QEC), current quadratic speedups will not fundamentally reveal the computational superiority of quantum techniques on realistic fault-tolerant devices [15]. This is caused by the practical limitations of quantum platforms, known as noisy intermediate-scales quantum (NISQ) devices [16]; here, the term "noisy" means that

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Chia-Hsiang Lin is with the Department of Electrical Engineering and the Miin Wu School of Computing, National Cheng Kung University, Tainan 70101, Taiwan (e-mail: chiahsiang.steven.lin@gmail.com).

You-Yao Chen is with the Department of Electrical Engineering, National Cheng Kung University, Tainan 70101, Taiwan (e-mail: q36114027@gs.ncku.edu.tw).

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qubits are easily affected by uncontrolled factors, while the term "intermediate-scales" refers to the limited available scale of the near-term quantum computers. Though achieving moderate image processing tasks with currently limited quantum facilities is possible [17], the hardware requirements for implementing large-scale quantum HRS algorithms may be beyond the ability of the current quantum platforms, which suffer from decoherence and fidelity issues, and the error occurs as the internal qubits lose their energy to the environment when being processed by the quantum logic gates [18], [19]. However, we still believe that quantum computing will serve as the key toward next-generation remote sensing technologies due to its strong potential in addressing NP-hard remote sensing problems like Craig simplex identification [2], [20], and it is expected to induce various revolutionary applications in other scientific fields as long as the QEC issue finds its groundbreaking advances.

Nevertheless, though there are various related research fields (e.g., quantum networks, quantum encrypted communication, and so on), one of the bottlenecks in introducing quantum computing into remote sensing is that the number of entangled quantum bits (qubits) is very limited (about 100) even in the well-known quantum computer nowadays [21], where the number of qubits entangled at the same time may even be much lower in the aforementioned NISQ computer; note that the famous quantum computer "International Business Machines Corporation (IBM) Eagle" has 127 qubits. Let us concisely recall the historical development beginning in 1969 when Stephen Wiesner proposed the first quantum computer. Feynman [22] described an idea of quantum mechanics devices for solving the issue of exponentially growing resources required when the instance size grows linearly. Deutsch [23] established the Church–Turing–Deutsch principle (CTD principle) to claim that every physical process can be simulated on the quantum mechanical version of a Turing machine. Venegas-Andraca and Bose proposed the quantum image storage and retrieval in a multiparticle quantum mechanical system [24], and Latorre proposed a new model for quantum image compression and storage [25]. It is interesting to remark that in late 2022, IBM announced another quantum computer "IBM Osprey" with 433 qubits, which is over three times more than the "IBM Eagle." One key factor here is how quantum mechanics allows two or more particles to exist in an entangled state, meaning that one of the particles in an entangled state can affect other particles even if they are far apart. Due to the limited number of entangled qubits, current quantum image processing technologies are either focus on those tasks just requiring very few output qubits (e.g., image classification) or on those tasks requiring just basic quantum gates operating on the quantum image state (e.g., image crop and rotation); other relatively simple image processing tasks of geometric and color transformation, including image flipping and color complement, can also be done via quantum computing [26], [27]. At this moment, advanced image processing tasks (e.g., image restoration) requiring a large number of qubits have never been seen in the literature, and considering the aforementioned large data volume in the typical hyperspectral image (HSI, whose representation requires a huge number of qubits), it is quite natural to ask the following question: *Is it feasible to achieve more advanced tasks in processing remotely sensed HSI (e.g., HSI restoration) using current quantum computing facilities?* If we can successfully demonstrate such feasibility as our step, it will be the milestone for inducing the subsequent development of all kinds of quantum remote sensing technologies (probably requiring fewer qubits) in the future.

As the limited number of qubits is insufficient to represent and process the images, exploring complementary techniques is needed. We have noticed ongoing works on quantum compression [28], but the technique uses classical deep learning to process quantum computing results; however, what we need in this work is to directly implement a deep network with quantum components for high-dimensional data quantum processing. Typically, a deep network first transforms/encodes the data into highly compressed features, which are then processed by the backbone network and mapped to the target features, followed by transforming/decoding back to the original data space; in other words, deep learning processes the highly compressed data in the feature space using much fewer bits [29]. So, it is quite natural to further incorporate deep learning into quantum remote sensing. In fact, deep learning has spurred numerous amazing applications since the AlphaGo [30] was proven to be able to mimic the best human Go players; some effective techniques include deep residual learning for image recognition [31], the Unet for image segmentation [32], the generative adversarial networks for unsupervised learning [33], the teacher–student architecture for knowledge distillation [34], as well as the deep $\boldsymbol{Q}$-quadratic small data learning for hyperspectral inverse imaging and remote sensing [29]. On the other hand, some deep learning theories (e.g., quantum convolution) have also been proposed to pave the way to extracting the quantum features [35], [36]. Thus, by deep learning, the image can be transformed into a feature space wherein one can use much fewer qubits to represent the features of the image. As in the feature space, the image has been compressed into much lower dimensional encodings, current quantum computer facilities are sufficient to process the deep image features, and then the processed quantum features will be measured yielding (collapsed) quantum image state that needs to be transformed back to the original image space, from which another challenging issue arises.

Even if the quantum image state has been well processed to the quantum state of the target image (QSTI), it cannot be perfectly retrieved as the QSTI will collapse to some eigenstate once it is measured. From this perspective, introducing deep learning into quantum HRS has brought us another great advantage. Specifically, previous quantum state retrieval requires a large number of quantum measurements to approximate the unobservable state, but even so, one still has no way to precisely obtain the QSTI. To solve the abovementioned challenges, we propose a quantum deep neural network to encode the hyperspectral information using very few qubits, as well as to learn the relation between the measuring statistics (associated with the collapsed-QSTI) and the target image (instead of directly retrieving the unob-

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servable QSTI). This work thereby makes a breakthrough to successfully reconstruct/restore the National Aeronautics and Space Administration's (NASA), Washington, DC, USA, damaged hyperspectral images using the proposed quantum deep network, and hence, achieves advanced quantum image processing tasks for the first time.

We concisely illustrate how we achieve so hereinafter. As clearly indicated in the very recent literature in *Nature Physics* [37], given all kinds of restrictions (e.g., a limited number of available qubits) in near-term quantum computers, we must need to explore some complementary techniques to achieve advanced quantum computing. The first complementary technique we proposed here is to design a deep encoder to highly compress the hyperspectral image of large data volume into much lower dimensional encodings, which can then be processed using our trickily designed hyperspectral quantum deep network (HyperQUEEN) architecture. The proposed HyperQUEEN purposely links different feature maps to generate more abstract quantum features, from which a fully connected quantum entanglement module is proposed to obtain the desired quantum state corresponding to the restored image features (i.e., the target quantum features). Then, to mimic the nonlinear pooling operator in deep learning, quantum measurements are performed just on half of the qubits to obtain the corresponding (collapsed) quantum state of the restored image features. Our mathematical analysis also guides us to incorporate a low-rank architecture with skip connections to avoid the gradient vanishing (GV) issue, which easily occurred in quantum deep learning [38]. Finally, besides the provable full expressibility (FE) of all unitary quantum operators in the proposed HyperQUEEN, we also employ the affine geometry nature of the hyperspectral data to design a low-rank decoder, which nonlinearly maps the collapsed quantum states (of the target/restored quantum features) back to the original image space, thereby successfully solving the highly ill-posed inverse problem of blindly restoring damaged hyperspectral images via HyperQUEEN.

The key contributions of the proposed HyperQUEEN technology are summarized as follows.

1) A radically new framework is developed under the currently immature quantum facilities (e.g., limited qubits), aiming at demonstrating the feasibility of achieving advanced quantum computing tasks in HRS. Our experiments demonstrate that hundred-scale qubits would be sufficient for HyperQUEEN to achieve challenging quantum computing tasks, well echoing the developing trends (e.g., IBM's Osprey). For the first time, HyperQUEEN solves the barren plateaus (BPs) effect [see. Theorem 1 and the quantum collapse (QC) effect for high-dimensional images (large data volume), so that it can read out a complete quantum image without distortion caused by QC. Note that even if the quantum image state can be well processed to the QSTI, it cannot be perfectly retrieved as the QSTI will collapse to some eigenstate once it is measured. This is the first work making a breakthrough to successfully achieve advanced high-dimensional quantum image processing tasks, so there is no reference for our design, meaning that HyperQUEEN is designed from scratch.
2) HyperQUEEN is a hybrid method blending classical and quantum deep learning. Due to the judiciously designed mechanisms for saving qubits (i.e., compression module f_ψ, and the low-rank module E_r), one major advantage of HyperQUEEN is that it is able to output a complete high-dimensional image (large data volume), while previous hybrid methods can only output quantities of much smaller size (e.g., classification [39]). This allows us to address many more real-world satellite image processing tasks.
3) For the first time, the theoretically guaranteed quantum FE (see. Theorem 2) enables the core quantum module E_ω in HyperQUEEN to successfully handle advanced quantum image processing tasks (e.g., inpainting, denoising), which are far more complicated than those classification-level tasks addressed in previous hybrid methods. For example, HyperQUEEN can solve tensor completion problems blindly without using any information of damaged positions, for blindly achieving fast quantum hyperspectral image inpainting on NASA's real-world satellite data. HyperQUEEN exhibits its superior performance even when compared with those nonblind methods. This means a lot for the upcoming quantum space era.
4) This work also serves as informative guidance (e.g., avoiding the over-entanglement design and so on) with clearly reported design philosophy for future investigators to adopt the proposed quantum deep learning framework to solve other challenging inverse problems.

As a side contribution, we have provided a concise but comprehensive background recall about quantum computing and quantum deep learning using basic linear algebra language, as well as a simple graphical illustration of QC, which are friendly for the remote sensing and signal processing people to start working on this exciting research line.

The remaining parts of this article are organized as follows. Some mathematical prerequisites of quantum computing are recalled in Section II, where the proposed HyperQUEEN architecture and related mathematical analysis are presented as well. In Section III, we demonstrate the practical applicability of HyperQUEEN by applying it to solve a challenging ill-posed HRS problem called blind hyperspectral inpainting. In Section IV, extensive experiments are conducted, showing that our quantum deep theory and architecture do successfully solve advanced quantum computing tasks in HRS. Finally, conclusions are drawn in Section V, while mathematical details are collectively presented in the Appendix.

Some standard notations are summarized here for better readability. $\mathbb{R}^n$ is the set of all the real-valued n-dimensional vectors; $\mathbb{R}^{m\times n}$ is the set of all the real-valued $(m \times n)$-dimensional matrices. $\mathbb{C}^n$ is the set of all the complex-valued n-dimensional vectors, and $\mathbb{C}^{m\times n}$ is the set of all the complex-valued $(m \times n)$-dimensional matrices. $U(N)$ denotes the set of unitary operations over $\mathbb{C}^N$. $[\boldsymbol{v}]_i$ is the ith entry of the vector $\boldsymbol{v}$, and $\boldsymbol{I}$ is the identity matrix. $\nabla(\cdot)$ denotes the gradient operator. $\boldsymbol{E}(\cdot)$, $\boldsymbol{Var}(\cdot)$ and $\boldsymbol{Cov}(\cdot,\cdot)$ denote the expectation, variance and covariance, respectively. The superscripts T, $\star$, and $\dagger$, respectively, denote the transpose, conjugate, and

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conjugate transpose. $\|\cdot\|$ and $\|\cdot\|_F$ are the Euclidean norm and Frobenius norm, respectively. Beside commonly seen notations above, other notations and some required preliminaries in quantum computing [40], [41] will be introduced in Section II.

II. Theoretical Development and HyperQUEEN Architecture for Quantum Deep Hyperspectral Remote Sensing

As discussed, quantum science is of great potential to lead future developments of remote sensing technologies, surely including HRS. We first recall some mathematical prerequisites in Section II-A, where Bloch sphere and some related concepts on quantum computing will be concisely presented to help readers smoothly access this article. Our geometry perspective to interpret the QC effect will also be shared in Section II-A

In Section II-B, quantum deep learning, as well as an effect related to the well-known GV in classical deep learning, will be discussed. Mathematical analysis guiding our design toward the low-rank quantum deep network will then be reported in Section II-C, where we derive and report some key principles/architectures behind the development of hyperspectral quantum technology for guiding future researchers working on the quantum deep HRS topic.

A. Dirac Notation System, Quantum Measurement, and Quantum Collapse Effect

This subsection recalls some concepts and mathematical definitions, including quantum states, quantum gates, quantum entanglement, and quantum measurements, in simple linear algebra language. The contents hereinafter also serve as a concise review for future investigators to quickly facilitate related research.

In the Dirac notation system, the "ket" and "bra" in quantum state representation correspond to column and row vectors in linear algebra, respectively. Given a Hilbert space $\mathcal{H}$, a vector $\psi \in \mathcal{H}$ is denoted by $|\psi\rangle$ and is referred to as a ket, which is represented as a column vector by choosing the orthonormal vector basis $\{|0\rangle, |1\rangle\}$, where the computational basis states are defined as $|0\rangle = [1, 0]^T$ and $|1\rangle = [0, 1]^T$. Then, every element $\psi \in \mathcal{H}$ can be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

for some $\alpha, \beta \in \mathbb{C}$. Accordingly, the bra $\langle\psi|$ is defined as the Hermitian of $|\psi\rangle$, which is the row vector

$$\langle\psi| = \alpha^\star\langle 0| + \beta^\star\langle 1|$$

where $\star$ denotes the conjugate operator, $\langle 0| = [1, 0]$ and $\langle 1| = [0, 1]$. It is interesting to remark that $\langle\psi|$ and $|\psi\rangle$ are called "bra" and "ket," respectively, because when they are combined, a "braket"-like notation $\langle\psi\|\psi\rangle$ is formed. In fact, the inner product of the two qubit states $|x\rangle$ and $|y\rangle$ can be conveniently denoted as $\langle x|y\rangle$, while their outer product is denoted as $|x\rangle\langle y|$.

The Dirac bra-ket notation system encodes the information using qubits string composed of $|0\rangle$ and $|1\rangle$, analogous to bit strings composed of 0 and 1 in classical computers. Different from the classical concept of a bit, a qubit can be in the superposition state (1), i.e., $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$; here, the quantum state $|0\rangle$ is of probability $|\alpha|^2$, while the state $|1\rangle$ is of probability $|\beta|^2$. As the probability amplitude $\alpha, \beta \in \mathbb{C}$ must satisfy the sum-to-one constraint $|\alpha|^2+|\beta|^2 = 1$, the following change of variables is always valid: $\alpha = e^{i\delta}\cos(\theta/2)$, $\beta = e^{i(\delta+\phi)}\sin(\theta/2)$, where $i = \sqrt{-1}$, $\theta \in [0, \pi]$, and $\delta, \phi \in [0, 2\pi]$. Thus, the qubit state can be reformulated as

$$|\psi\rangle = e^{i\delta}\left(\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle\right)$$

for which $e^{i\delta}$ and $e^{i\phi}$ are referred to as the global phase and the relative phase, respectively. As will be detailed later, a measurement result is of the form $\langle\psi|\boldsymbol{H}|\psi\rangle$ for some Hermitian matrix $\boldsymbol{H}$, implying that the global phase is never reflected in the measurement result, and hence, ignorable. In other words, a qubit state can be parameterized by (θ, ϕ) and represented using the Bloch sphere as illustrated in Fig. 1. For example, we have $(\theta, \phi) = (\pi, 0)$ for the qubit state $|1\rangle$, which is, thus, placed at the bottom (corresponding to $\theta = \pi$) of the Bloch sphere.

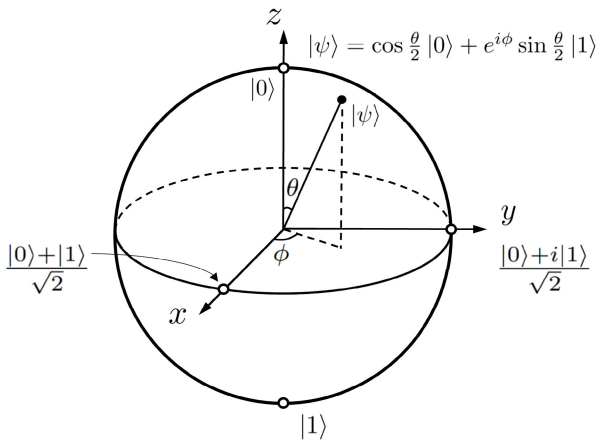
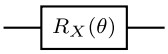
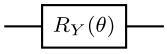
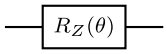
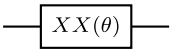
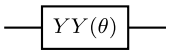
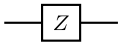
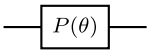
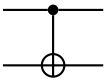
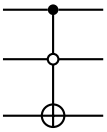
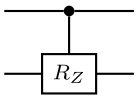


Fig. 1. Qubit state $|\psi\rangle$ in the quantum computer can be parameterized by $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$ in a Bloch sphere.

Analogous to logical gates in classical computers, quantum computers are built based on quantum gates [42]. Some frequently adopted quantum gates, together with their symbols and mathematical definitions, are summarized in Table I. Since a quantum gate is implemented via a Hamiltonian for a specific time, yielding a unitary time evolution according to the Schrödinger equation, the corresponding quantum operator is unitary, and thus, linear [41]. For example, in Table I, some basic rotation gates (e.g., rotation X) and their variations (e.g., Ising XX, controlled rotation, and so on), as well as phase shift and Pauli-Z gates, are all unitary. We remark that, unlike classical computers wherein the gates are mostly irreversible, all the quantum gates are reversible due to the unitary property. Very few gates (e.g., NOT gate) share the same form in both classical and quantum computers. The NOT gate, also known as Pauli-X gate, can be extended to the controlled NOT (CNOT) when the controlled qubit state is $|1\rangle$, and to the controlled CNOT (CCNOT) when the controlled qubit state is $|10\rangle$, as detailed in Table I. It is important to remark that if a finite set $\mathcal{S}$ can generate or

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TABLE I
FREQUENTLY USED QUANTUM GATES AND THEIR MATHEMATICAL DEFINITIONS [41], [44]

Quantum Gate	Symbol	Unitary Operator
Rotation X		$\begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$
Rotation Y		$\begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$
Rotation Z		$\begin{pmatrix} e^{-i(x/2)} & 0 \\ 0 & e^{i(x/2)} \end{pmatrix}$
Ising XX		$\begin{pmatrix} \cos(\theta/2) & 0 & 0 & -i\sin(\theta/2) \\ 0 & \cos(\theta/2) & -i\sin(\theta/2) & 0 \\ 0 & -i\sin(\theta/2) & \cos(\theta/2) & 0 \\ -i\sin(\theta/2) & 0 & 0 & \cos(\theta/2) \end{pmatrix}$
Ising YY		$\begin{pmatrix} \cos(\theta/2) & 0 & 0 & i\sin(\theta/2) \\ 0 & \cos(\theta/2) & -i\sin(\theta/2) & 0 \\ 0 & -i\sin(\theta/2) & \cos(\theta/2) & 0 \\ i\sin(\theta/2) & 0 & 0 & \cos(\theta/2) \end{pmatrix}$
Pauli-Z		$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Phase Shift		$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$
NOT		$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Controlled NOT (CNOT, with controlled state $\|1\rangle$)		$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$
Controlled CNOT (CCNOT, with controlled state $\|10\rangle$)		$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$
Controlled Rotation Z		$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{-i(\theta/2)} & 0 \\ 0 & 0 & 0 & e^{i(\theta/2)} \end{pmatrix}$

closely approximate any other unitary operations in a finite sequence (e.g., $\mathcal{S} := \{R_X, R_Y, R_Z, P, \text{CNOT}\}$), the set $\mathcal{S}$ is called a universal quantum gate set (UQGS), mathematically meaning that $\mathcal{S}$ can generate a topologically dense subset of the projective group of 2×2 unitaries [43].

Surely, quantum computers can process multiple qubits, which can be represented using the tensor product of multiple computational basis states. In the n-qubit system, the computational basis state $|10\ldots0\rangle_n$ can be defined using the tensor product $\otimes$ (a.k.a. Kronecker product in linear algebra) as

$$|10\ldots0\rangle_n = |1\rangle \otimes |0\rangle \ldots \otimes |0\rangle = |1\rangle \otimes |0\rangle^{\otimes n-1}$$

which can also be conveniently denoted as $|10\ldots0\rangle_n = \left|2^{n-1}\right\rangle_n$, where the number n of qubits may not be explicitly specified if it is clear from the context. Accordingly, if $U_1 \ldots U_n$ act on the n qubits $|x_1 x_2 \ldots x_n\rangle$, respectively, the overall quantum operator U can be expressed as

$$U = U_1 \otimes U_2 \otimes \ldots \otimes U_n. \tag{2}$$

The other computational basis states of the n-qubit system can also be defined in a similar manner, leading to the following superposition representation of a general state $|\psi\rangle$ in the n-qubit system, that is,

$$|\psi\rangle_n = \sum_{k=0}^{2^n-1} \alpha_k |k\rangle \tag{3}$$

where the probability amplitudes $\alpha_0, \ldots, \alpha_{2^n-1} \in \mathbb{C}$ satisfy $\sum_{k=0}^{2^n-1} |\alpha_k|^2 = 1$; similarly, we have $\langle\psi| = \sum_{k=0}^{2^n-1} \alpha_k^\star \langle k|$. If a pure state $|\psi\rangle_n$ [i.e., a state expressible by (3)] can be further decomposed into subsystems $|\psi\rangle_n = |\phi_1\rangle_m \otimes |\phi_2\rangle_\ell$ with $n = m + \ell$, then there is no *quantum entanglement* between subsystems. Assuming that the pure state $|\psi_s\rangle$ occurs with probability p_s due to some noise effect, the corresponding density matrix is the convex combination $\rho = \sum_s p_s |\psi_s\rangle\langle\psi_s| \in \mathbb{C}^{2^n \times 2^n}$, which is a Hermitian operator of trace one [41].

Unlike classical computers, a state $|\psi\rangle$ in a quantum computer may not be perfectly retrieved. Usually, a Hermitian measurement operator H is used to obtain a collapsed version

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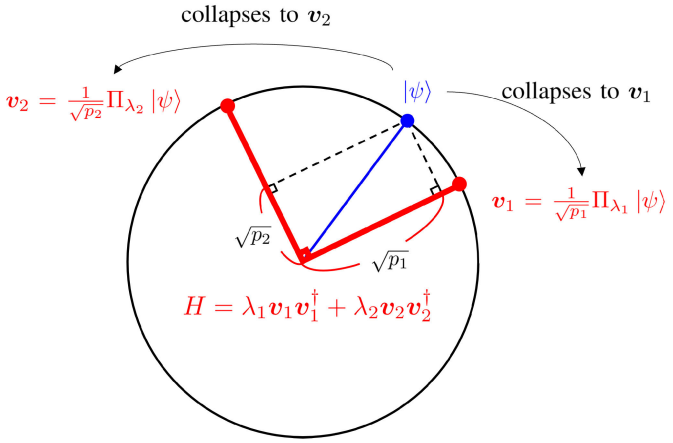


Fig. 2. Geometry perspective to QC. When the quantum state $|\psi\rangle$ is measured by the Hermitian operator $H := \sum_{i=1}^{2}\lambda_i\Pi_{\lambda_i} = \sum_{i=1}^{2}\lambda_i(\boldsymbol{v}_i\boldsymbol{v}_i^\dagger)$ (with $\boldsymbol{v}_i$ being the ith eigenvector of H), the state $|\psi\rangle$ will collapse to the eigenstate $\boldsymbol{v}_i = (1/(p_i{}^{1/2}))\Pi_{\lambda_i}|\psi\rangle$ with probability $p_i = \langle\psi|\Pi_{\lambda_i}|\psi\rangle$, for $i = 1, 2$. The depicted geometry meaning of the probabilities $\{p_1, p_2\}$ satisfies the Pythagorean theorem, i.e., $((p_1)^{1/2})^2 + (\sqrt{p_2})^2 = \||\psi\rangle\|^2 = 1$.

of $|\psi\rangle$. To be specific, when the state $|\psi\rangle$ is measured using H, the measuring result will be some eigenvalue λ_i of H, and the probability of getting the measuring result λ_i is $p_i = \langle\psi|\Pi_{\lambda_i}|\psi\rangle$, where Π_{λ_i} is the projector onto the eigenspace corresponding to λ_i. Furthermore, when the measuring result is λ_i, the quantum state $|\psi\rangle$ will collapse to $(1/(p_i)^{1/2})\Pi_{\lambda_i}|\psi\rangle$, as illustrated in Fig. 2. As the quantum state $|\psi\rangle$ will collapse to some eigenstate once it is measured, one instead aims to retrieve some statistics (e.g., expectation) of the measuring results. Specifically, the expectation $\langle H\rangle$ of the measuring results with respect to (w.r.t.) the Hermitian measurement operator H can be derived as

$$\langle H\rangle = \sum_i p_i\lambda_i = \sum_i \lambda_i\langle\psi|\Pi_{\lambda_i}|\psi\rangle = \langle\psi|H|\psi\rangle \tag{4}$$

where the last equality follows from the eigenvalue decomposition (EVD) $H := \sum_i \lambda_i\Pi_{\lambda_i}$.

B. Quantum Deep Learning and Barren Plateaus Effect

As discussed in a number of contexts [45], [46], deep quantum networks of the following form are commonly employed:

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} U_\ell(\theta_\ell)W_\ell \tag{5}$$

where $U_\ell(\theta_\ell) = \exp(-i\theta_\ell V_\ell)$ is a learnable unitary operator, W_ℓ is a nonlearnable unitary operator (e.g., identity matrix), V_ℓ is some Hermitian operator, and $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_L]$ is the trainable network parameter vector. As reported in the nature articles, (5) is actually a more general form of [46, eq. (1)] and is exactly the same as [38, eq. (1)]. The above-mentioned form is a natural choice due to a straightforward evaluation of the gradient w.r.t. most loss functions for training $\boldsymbol{\theta}$ [38]. By (4), the expectation of the corresponding quantum measurement w.r.t. the Hermitian operator $\boldsymbol{H}$ is set as the loss function $\widetilde{E}$, that is,

$$\widetilde{E}(\boldsymbol{\theta}) = \langle 0|U(\boldsymbol{\theta})^\dagger \boldsymbol{H} U(\boldsymbol{\theta})|0\rangle.$$

For analysis purposes, the quantum network is divided into two parts, i.e., $U_- = \prod_{\ell=1}^{k-1}U_\ell(\theta_\ell)W_\ell$ and $U_+ = \prod_{\ell=k}^{L}U_\ell(\theta_\ell)W_\ell$ and is assumed to have the property that for any gradient direction $\partial_k\widetilde{E} \equiv ((\partial\widetilde{E}(\boldsymbol{\theta}))/\partial\theta_k)$, the quantum network $U(\boldsymbol{\theta})$ is sufficiently random such that U_- and U_+ are independent and that either U_-, U_+, or both match the Haar distribution up to the second moment [38].

To recall the well-known BP effect [38], we need to define the concept of t-design. Assuming that the unitary operator V_i is sampled with probability p_i, the ensemble $\{p_i, V_i\}$ is called a t-design if

$$\sum_i p_i V_i^{\otimes t}\rho(V_i^\dagger)^{\otimes t} = \int d\mu(U)U^{\otimes t}\rho(U^\dagger)^{\otimes t} \tag{6}$$

where the integral is w.r.t. the Haar measure on the unitary group (i.e., $d\mu(U) = d\mu_{\text{Haar}}(U)$) [38].[1] There are two important results. First, if at least one of the independent networks U_- and U_+ forms 1-design, then the expectation value of the gradient vanishes, that is,

$$\boldsymbol{E}\left(\partial_k\widetilde{E}\right) = 0 \tag{7}$$

which will be assumed in our analysis. Second, if at least one of U_- and U_+ forms two-design, then an ansatz state on a BP will be produced with high probability giving no interesting search directions in sight, i.e., the variance $\boldsymbol{Var}(\partial_k\widetilde{E}) \approx (1/2^{zn})V_H$ decreases exponentially in the number of qubits n, where $z \in \{2, 3\}$ and V_H is a term depending on the Hermitian operator $\boldsymbol{H}$ [38]. Accordingly, we adopt the assumption of

$$\lim_{n\to\infty}\boldsymbol{Var}(\partial_k\widetilde{E}) \to 0 \tag{8}$$

in our subsequent analysis and have the following definition.

Definition 1: A quantum deep network is said to have the BP effect if $\boldsymbol{E}(\partial_k\widetilde{E}) \to 0$ and $\boldsymbol{Var}(\partial_k\widetilde{E}) \to 0$ as $n \to \infty$. □

The BP effect in quantum deep learning looks somewhat similar to the GV effect in classical deep learning, but BP is not caused by the far distance between the layers; in fact, existing quantum neural networks mostly have much fewer layers when compared with the classical ones. The BP effect plays an essential role in quantum deep learning as will be proved next.

C. Theoretical Analysis and Proposed Low-Rank Quantum Deep Network

We are now ready to derive and report some key principles/architectures behind the development of hyperspectral

[1]This definition is to avoid the unrealistic requirement of exponential resources in the quantum circuits for satisfying the unique left- and right-invariant measuring properties of the Haar measure on the unitary group, that is,

$$\int d\mu(U)f(U) = \int d\mu(U)f(VU) = \int d\mu(U)f(UV)$$

for any $f(U)$ (e.g., a polynomial of at most degree t in the matrix elements of U) and $V \in U(N)$, where $U(N)$ is set as the integration domain (i.e., the set of unitary operations over $\mathbb{C}^N$) when not explicitly specified. This motivates the concept of unitary t-design, which satisfies the above-mentioned invariance property (needed when analyzing the BP effect [38]) for restricted classes of $f(U)$ while often requiring just polynomial resources. Simply speaking, the definition (6) says that averaging over the t-design $\{p_i, V_i\}$ is equivalent to averaging over the unitary group w.r.t. the Haar measure [38].

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quantum technology for guiding future researchers working on this topic.

Let us begin by recalling the affine geometry property of a typical M-band hyperspectral image (HSI) $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L] \in \mathbb{R}^{M\times L}$, where $\boldsymbol{x}_n \in \mathbb{R}^M$ is the nth hyperspectral pixel. It is known that the pixels in a given HSI often concentrate on a very low-dimensional subspace of $\mathbb{R}^M$ [47]. Thus, there exists $N \ll M$, as well as the corresponding basis $\boldsymbol{E} \in \mathbb{R}^{M\times N}$, such that each pixel $\boldsymbol{x}_n$ can be represented as $\boldsymbol{x}_n = \boldsymbol{E}\boldsymbol{s}_n$ for some coefficient vector $\boldsymbol{s}_n \in \mathbb{R}^N$, thereby leading to the following low-rank representation, that is:

$$\boldsymbol{X} = \boldsymbol{E}\boldsymbol{S} := E_r(\boldsymbol{S}) \tag{9}$$

where $\boldsymbol{S} \in \mathbb{R}^{N\times L}$ is the coefficient matrix, and E_r is a module/function to model the low-rank property. As the basis $\boldsymbol{E}$ is often obtained by the first N principal eigenvectors of the mean-removed data-covariance matrix [48], the N-band multispectral image $\boldsymbol{S} \in \mathbb{R}^{N\times L}$ is also referred to as the eigenimage [49]. The model order is empirically set as the default value $N := 8$, which yields good performances for all four representative landscapes (i.e., city, mountain, farm, and lake).

Inspired by the great success of convolutional neural network (CNN), wherein both the shared-weight mechanism and partly connected layers aim at reducing the network parameters, we reduce the number of parameters in the hyperspectral quantum network by designing a low-rank quantum network architecture based on (9). Specifically, the first half of our quantum network should endeavor to output the N-band multispectral image $\boldsymbol{S}$ (corresponding to some target HSI $\boldsymbol{X}$), while the second half should perform spectral upsampling capturing the effect of $\boldsymbol{E}$ to return back to the hyperspectral space, as illustrated in Fig. 3(a). To understand why this architecture can save a great number of parameters, we first notice that the number of parameters in $\boldsymbol{X}$ is ML, while the number of parameters in $(\boldsymbol{E}, \boldsymbol{S})$ is $MN+NL$; note that $MN+NL \ll ML$ as we often have $N \ll M \ll L$ in typical HSIs. Intuitively, performing spectral upsampling [via the second module in Fig. 3(a)] after estimating the spatial details [via the first module in Fig. 3(a)] greatly saves the number of parameters, when comparing with the approach of directly learning the spatial details in the hyperspectral domain. Furthermore, since $\boldsymbol{E}$ is formed by some eigenvectors, it is essentially (or can be augmented to) a unitary matrix, thereby allowing a quantum implementation of this low-rank quantum network architecture (see. Section II-A). Implementation of $\boldsymbol{E}$ is relatively simple as will be seen in Section III, so next we will focus on the architecture design of the $\boldsymbol{S}$-module [i.e., the first module in Fig. 3(a)].

The first challenge in designing the $\boldsymbol{S}$-module lies within the very limited number of qubits in current quantum computers. To address this issue, we propose to highly compress the input HSI (i.e., HSI to be processed), followed by performing image processing just in the compressed data domain wherein much fewer qubits are required. This approach echoes the current trend in the Starlink of miniaturized satellites, for which the storage space is highly restrictive thus forcing

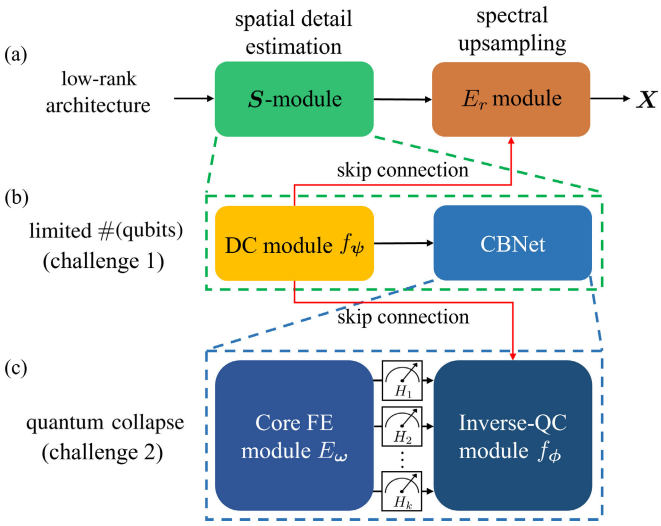


Fig. 3. Proposed HyperQUEEN architecture [i.e., $E_r(f_{\boldsymbol{\phi}}(E_{\boldsymbol{\omega}}(f_{\boldsymbol{\psi}}(\cdot))))$] and design principles here serve as a guide for future researchers working on quantum deep HRS. (a) Low-rank quantum network architecture employs the hyperspectral subspace phenomenon to reduce the number of learning parameters, thereby increasing the stability and efficiency, where E_r is essentially a unitary operator that hence can be implemented by suitable quantum gates. (b) DC module $f_{\boldsymbol{\psi}}$ is used to highly compress the input image into very low-dimensional encodings/features that can be represented using much fewer qubits, followed by processing the data via CBNet directly in the compressed data domain, well in line with the trend of Starlink of miniaturized satellites. (c) CBNet not only needs to ensure the FE of all quantum unitary operators via the core FE module $E_{\boldsymbol{\omega}}$ but also judiciously introduces an inverse-QC module $f_{\boldsymbol{\phi}}$ to compensate for the missing information caused by the QC effect.

related signal processing tasks to be done directly in the compressed data domain. Though having different purposes, the above-mentioned idea is similar to the encoding part in deep learning, which is often nonlinear, and thus, suitable to achieve the high compression ratio required for current quantum computers. All in all, as illustrated in Fig. 3(b), the first part of $\boldsymbol{S}$-module employs a nonlinear deep encoder to highly compress the input HSI into the quantum feature space. This deep compression (DC) module, denoted as $f_{\boldsymbol{\psi}}$, then allows the core backbone network [CBNet, i.e., the second part of $\boldsymbol{S}$-module; see. Fig. 3(b)] to focus on processing the low-dimensional encodings/features that can be represented using much fewer qubits.

The most critical part would be the design of CBNet, which should solve another key challenge made by the QC effect. Specifically, even if the quantum image state can be well processed to the QSTI, it cannot be perfectly retrieved as the QSTI will collapse to some eigenstate once it is measured; see Fig. 2. Therefore, instead of directly retrieving the unobservable QSTI, another key principle is to design a module to learn the inverse QC mapping, $f_{\boldsymbol{\phi}}$, from some measuring statistics (associated with the collapsed-QSTI) to the target image. The statistics here refer to some quantum measurements in the form of (4), where we try to observe the QSTI from multiangles using k distinct Hermitian measurement operators $H_1, \ldots, H_k$, as illustrated in Fig. 3(c).

Another key module $E_{\boldsymbol{\omega}}$ in CBNet is to conduct the core processing in the quantum feature space. To be precise, the

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quantum neural network $E_{\boldsymbol{\omega}}(\boldsymbol{x})$ can be explicitly written as

$$E_{\boldsymbol{\omega}}(\boldsymbol{x}) = \begin{pmatrix} E_1(\boldsymbol{x},\boldsymbol{\omega}) \\ E_2(\boldsymbol{x},\boldsymbol{\omega}) \\ \vdots \\ E_k(\boldsymbol{x},\boldsymbol{\omega}) \end{pmatrix} = \begin{pmatrix} \langle 0|Z^\dagger(\boldsymbol{x},\boldsymbol{\omega})H_1 Z(\boldsymbol{x},\boldsymbol{\omega})|0\rangle \\ \langle 0|Z^\dagger(\boldsymbol{x},\boldsymbol{\omega})H_2 Z(\boldsymbol{x},\boldsymbol{\omega})|0\rangle \\ \vdots \\ \langle 0|Z^\dagger(\boldsymbol{x},\boldsymbol{\omega})H_k Z(\boldsymbol{x},\boldsymbol{\omega})|0\rangle \end{pmatrix}$$

where $\boldsymbol{\omega}$ is the training parameter, $\boldsymbol{x}$ is the training embedding, $E_j(\boldsymbol{x},\boldsymbol{\omega}) := \langle 0|Z^\dagger(\boldsymbol{x},\boldsymbol{\omega})H_j Z(\boldsymbol{x},\boldsymbol{\omega})|0\rangle$ is the jth quantum measuring result of $|\psi\rangle$ according to (4), and here, the quantum state $|\psi\rangle := Z(\boldsymbol{x},\boldsymbol{\omega})|0\rangle$ is exactly the aforementioned QSTI obtained using the quantum network (see Section II-B)

$$Z(\boldsymbol{x},\boldsymbol{\omega}) := \prod_i Z_i(\omega_i)\tilde{W}_i \prod_j V_j(x_j)\hat{W}_j$$

in which $\omega_i := [\boldsymbol{\omega}]_i$, $x_j := [\boldsymbol{x}]_j$, Z_i is learnable unitary operators, V_j is unitary operators for embedding the data $\boldsymbol{x}$, and $\tilde{W}_i$ and $\tilde{W}_j$ are nonlearnable unitary operators like identity or CCNOT; see (5) [38], [45], [46]. By grouping $\boldsymbol{x}$ and $\boldsymbol{\omega}$ into $\boldsymbol{\theta} = [\boldsymbol{x}^T, \boldsymbol{\omega}^T]^T$ and thereby reformulating $Z(\boldsymbol{x},\boldsymbol{\omega})$ into $U(\boldsymbol{\theta}) = \prod_\ell U_\ell(\theta_\ell)W_\ell$, where U_ℓ and W_ℓ are the learnable and nonlearnable unitary operations, respectively, the core FE module $E_{\boldsymbol{\omega}}$ in Fig. 3(c) can be concisely reformulated as

$$E_{\boldsymbol{\omega}}(\boldsymbol{x}) = \begin{pmatrix} \langle 0|U^\dagger(\boldsymbol{\theta})H_1 U(\boldsymbol{\theta})|0\rangle \\ \langle 0|U^\dagger(\boldsymbol{\theta})H_2 U(\boldsymbol{\theta})|0\rangle \\ \vdots \\ \langle 0|U^\dagger(\boldsymbol{\theta})H_k U(\boldsymbol{\theta})|0\rangle \end{pmatrix} \equiv \begin{pmatrix} E_1(\boldsymbol{x},\boldsymbol{\omega}) \\ E_2(\boldsymbol{x},\boldsymbol{\omega}) \\ \vdots \\ E_k(\boldsymbol{x},\boldsymbol{\omega}) \end{pmatrix}. \quad (10)$$

The most critical design principle here is to ensure its FE, i.e., the ability to express all kinds of unitary operators by $E_{\boldsymbol{\omega}}$, such that the proposed quantum network can be trained to realize any desired quantum operator.

The above-mentioned discussions are collectively summarized in Fig. 3, where the overall HyperQUEEN architecture can be mathematically written as $E_r(f_{\boldsymbol{\phi}}(E_{\boldsymbol{\omega}}(f_{\boldsymbol{\psi}}(\cdot))))$. Before applying HyperQUEEN to solve challenging HRS problems (see Section III), we analyze its behavior during the training phase, as stated and proved in the following theorem.

Theorem 1: Assuming the Lipschitz decoding scheme [i.e., $\|\nabla_{E_{\boldsymbol{\omega}}}(E_r \circ f_{\boldsymbol{\phi}})\| \le \alpha$ for some constant α], independent measuring probes and t-design mechanisms with $t \in \{1, 2\}$, the outlined quantum deep architecture, i.e., the HyperQUEEN $E_r(f_{\boldsymbol{\phi}}(E_{\boldsymbol{\omega}}(f_{\boldsymbol{\psi}}(\cdot))))$, has the BP effect on the core module $E_{\boldsymbol{\omega}}$ [defined in (10)] with training parameter $\boldsymbol{\omega}$. □

The proof of Theorem 1 is relegated to Appendix A. This analysis tells us that when HyperQUEEN goes too deep (implying a large n), it would suffer from the GV effect during the training phase. So, avoiding too many layers and/or adding some skip connections among the modules would be good strategies when implementing the HyperQUEEN architecture, as illustrated in Section III.

III. Application of HyperQUEEN to Blind Hyperspectral Tensor Completion

In this section, we demonstrate how our HyperQUEEN theory can be applied to solve challenging ill-posed inverse problems. Specifically, following Fig. 3, we will design the modules in the HyperQUEEN architecture $E_r(f_{\boldsymbol{\phi}}(E_{\boldsymbol{\omega}}(f_{\boldsymbol{\psi}}(\cdot))))$ to blindly solve the quantum hyperspectral inpainting (QHI), a tensor completion problem. Since this work achieves the advanced quantum image processing tasks (e.g., QHI) for the first time, there is no reference for our design, meaning that the design reported below has been done from scratch.[2]

We remark that unlike the core FE module $E_{\boldsymbol{\omega}}$ and the low-rank module E_r, which can be implemented by quantum unitary gates (see Section II-C), the DC module $f_{\boldsymbol{\psi}}$ (for nonlinear compression) and the inverse-QC module $f_{\boldsymbol{\phi}}$ (for nonlinear inverse mapping) are both nonlinear and, thus, nonunitary. In other words, unlike those unitary modules (denoted by E), those nonlinear modules (denoted by f) are not implementable by quantum networks, because quantum gates are unitary and thus linear (see Section II-A and Table I). Therefore, we implement the DC module $f_{\boldsymbol{\psi}}$ and the inverse-QC module $f_{\boldsymbol{\phi}}$ using classical deep networks, with implementation details reported in Table II, where "CB(c, s, p)" [respectively, "TCB(c, s)"] denotes convolution block (respectively, transposed convolution block) composed of convolution Conv(c, s, p) [respectively, transposed convolution TConv(c, s)], batch normalization (BN) and LeakyReLU; here, LeakyReLU is of negative slope 0.2, and (c, s, p), respectively, specify the number of output channels, the kernel size of $s \times s$, and the number of paddings. Note that compared with LeakyReLU, other frequently seen activation functions like Sigmoid or Tanh easily cause the GV effect, as reported in [50, Sec. VI]. From the design in Table II, the DC module $f_{\boldsymbol{\psi}}$ highly compresses the hyperspectral data into very low-dimensional features/encodings with a compression ratio of about 0.018%, and the inverse-QC module (together with the low-rank module E_r to be designed later) does have the capability to decode the collapsed-QSTI to reconstruct high-fidelity hyperspectral data as will be experimentally demonstrated in Section IV.

The most critical part of HyperQUEEN is actually the quantum implementation of the core FE module $E_{\boldsymbol{\omega}}$, for which the deep quantum layers are reported in Table II, where the quantum gates "$R_Y(g, n)$," "$R_Z(g, n)$," "$XX(g, n)$," and "$Z(g, n)$" follow the symbols used in Table I, in which (g, n), respectively, denote the number of parallel groups and the number of qubits in each group. On the other hand, to capture the quantum entanglement effect, we customize a Toffoli entanglement module composed of four quantum gates of "CCNOT(cw_1,cw_2,w)", where (cw_1,cw_2,w) specify the three wires in CCNOT (see Table I) with cw_1

[2]The design of the core quantum part has been done by first identifying some potential architectures via empirical observations, followed by adjusting some potential candidates to be a symmetric one (e.g., the five-layer Ising-rotation architecture "$R_Y - XX - R_X - XX - R_Y$" to be introduced later) as well as to satisfy the provable quantum FE (see Theorem 2). The way we came up with the design would be helpful experience to guide future investigators.

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TABLE II
IMPLEMENTATION DETAILS OF HYPERQUEEN, WHERE "MAXPOOL" DENOTES THE MAX POOLING, AND THE UPSAMPLING LAYERS ARE IMPLEMENTED BY BILINEAR INTERPOLATION

	Layer	Configuration	Output Size
	Input	-	172×128×128
DC module f_{ψ}	ConvModule 1	CB(8,3,1)×1 CB(16,3,0)×2 2×2 MaxPool	16×62×62
	ConvModule 2	CB(32,3,0)×3 2×2 MaxPool	32×28×28
	ConvModule 3	CB(64,3,0)×4 2×2 MaxPool	64×10×10
	ConvModule 4	CB(128,3,0)×4	128×2×2
Core FE module E_{ω} (Theorem 2)	Reshape	-	128×4
	Data Embedding	$R_Z(128,4)$	128×4
	Unitary Gate 1	$R_Y(128,4)$	128×4
	Reshape	-	256×2
	Unitary Gate 2	$XX(256,2)$	256×2
	Reshape	-	128×4
	Unitary Gate 3	$R_X(128,4)$	128×4
	Reshape	-	256×2
	Unitary Gate 4	$XX(256,2)$	256×2
	Reshape	-	128×4
	Unitary Gate 5	$R_Y(128,4)$	128×4
	Toffoli Entanglement	CCNOT(0,1,2)	128×4
		CCNOT(1,2,3)	128×4
		CCNOT(2,3,0)	128×4
		CCNOT(3,0,1)	128×4
	QC Measurement	$Z(128,2)$	128×2
	Reshape	-	64×2×2
Inverse-QC module f_{ϕ}	TConvModule 1	TCB(64,3)×4 2×2 Up-sample	64×20×20
	TConvModule 2	TCB(64,3)×2 TCB(32,3)×2 2×2 Up-sample	32×56×56
	TConvModule 3	TCB(32,3)×2 TCB(16,3)×1 2×2 Up-sample	16×124×124
	TConvModule 4	TCB(16,3)×1 TCB(8,3)×1	8×128×128
Low-rank module E_r	E_r Module	CB(172,1,0)×1 Conv(172,1,0)×1	172×128×128

(respectively, cw_2) being the first control wire (respectively, second control wire) and w is the wire the operation acts on. In the proposed HyperQUEEN, the two control wires (cw_1,cw_2) are configured with control qubit states $|1\rangle$ and $|0\rangle$, respectively. The detailed deployment of the Toffoli entanglement module is given in Table II and graphically illustrated in Fig. 4. Other parts of the core FE module E_{ω}, including quantum angle encoding (data embedding), quantum FE gates and quantum measurements (see Fig. 4), are, respectively, illustrated in the following.

First, we employ the quantum angle encoding[3] Scheme [53] to transfer the information embedded in the compact features (output from the DC module f_{ψ}) into the core FE module E_{ω} for subsequent quantum processing. Specifically, each element x in the output feature vector is encoded as the angle of the quantum rotation Z gate [i.e., $R_Z(\theta)$] [54, Sec. II], as illustrated in Fig. 4. Mathematically, the angle encoding encodes the feature value x by implementing the unitary operator $R_Z(\theta)|_{\theta:=x}$, where $R_Z(\theta)$ has been defined in Table I. Second, before illustrating how we achieve the desired FE of all the quantum unitary gates, we explain how the well-processed quantum features can be fed into the inverse-QC module f_{ϕ}. This can be done by the quantum measurement module in Fig. 4, which can be implemented by two Pauli-Z quantum gates operated only on the qubits with odd indices; see the design in Table II. Measuring just those qubits with odd indices transfers 128 × 4 quantum features into 128 × 2 quantum features, as reported in Table II, and this quantum mechanism is analogous to the max pooling in classical deep learning; note that the selection of the measuring qubits (e.g., qubits with even indices) is not a key factor for the final performance. Also, as discussed in Section II-A, retrieving some statistics is more meaningful due to the QC effect; specifically, we estimate the expectation of the measuring result by averaging over 1000 Pauli-Z measurements, and these retrieved quantum statistics are then fed into the inverse-QC module f_{ϕ}, as illustrated in Fig. 4. With the above-mentioned specified input–output mechanisms of the core FE module E_{ω}, the remaining task is to design the quantum gates for achieving the most critical FE to express all possible quantum unitary operators through E_{ω}.

To achieve FE quantum gates, the five-layer Ising-rotation architecture "$R_Y - XX - R_X - XX - R_Y$" is designed with detailed configurations specified as the Unitary Gates 1–5 in Table II. The Unitary Gates 2 and 4 are purposely designed to across qubits, not only for introducing Ising entanglement but for increasing the communications among qubits, as illustrated in Fig. 4. The FE of the proposed five-layer Ising-rotation architecture can be guaranteed by the following theorem.

Theorem 2: Given any unitary operator U_k acting on the kth qubit for $k \in \{0,1,2,3\}$, yielding the overall quantum unitary operator $U := \bigotimes_{k=0}^{3} U_k$ [see (2)], there exist real-valued $\{\alpha_k, \beta_k, \gamma_k, \delta_k, \theta_k\}$ such that

$$U = AI_0(\theta_0,\theta_1)BI_1(\theta_2,\theta_3)C$$

where $A \triangleq \bigotimes_{k=0}^{3} e^{i\alpha_k} R_Y(\beta_k)$, $B \triangleq \bigotimes_{k=0}^{3} R_X(\gamma_k)$, and $C \triangleq \bigotimes_{k=0}^{3} R_Y(\delta_k)$ are the overall quantum effects of the three rotation layers, while $I_0(\theta_0,\theta_1) \triangleq XX_{03}(\theta_0) \otimes XX_{12}(\theta_1)$ and $I_1(\theta_2,\theta_3) \triangleq XX_{01}(\theta_2) \otimes XX_{23}(\theta_3)$ are the overall quantum effects of the two Ising entanglement layers; note that $XX_{mn}(\theta)$ here means that the Ising XX gate $XX(\theta)$ is operated on the mth and nth qubits, and all the other quantum gates involved here are defined in Table I. □

Theorem 2 guarantees the FE of the proposed five-layer Ising-rotation architecture, where $e^{i\alpha_k}$ may be viewed as the global phase (thus, ignorable; see Section II-A), ensuring that there exist some trainable quantum network parameters to realize any desired quantum unitary operators, as proved in Appendix B. Considering that the great success of deep learning consists in its capability of merging different features of previous layers to extract more abstract features for deeper

[3]Under quantum angle encoding, the 128 × 4 quantum features are represented by 512 qubits, and this hundred-scale simulation can be implemented on the Pytorch and PennyLane (the leading tool for programming quantum computers) platforms [51], [52]. This scale is manageable by the GPU memory of our device; note that both the computational time and memory usage grow exponentially as the number of qubits increases. This scale makes sense in practice, as IBM has achieved a hundred-scale qubit system (IBM Osprey) in 2022 and claimed a 4000-qubit system in 2025.

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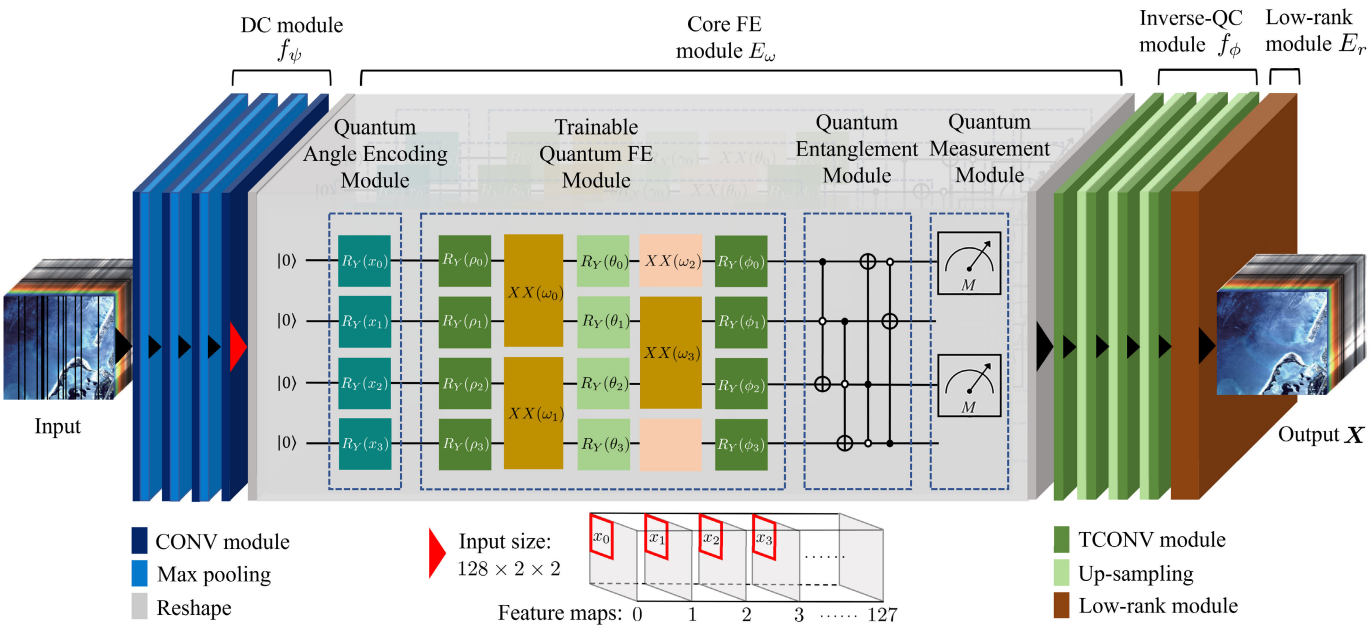


Fig. 4. Schematic of the implementation of HyperQUEEN, where the learning parameters include $(\beta_i, \gamma_i, \delta_i, \theta_i)$ for all $i \in \{0, 1, 2, 3\}$; note that $x_0, \ldots, x_3$ (not learning parameters) are for data encoding. The central core FE module E_ω (with a circuit depth of 10) is composed of quantum components, while the other modules (f_ψ, f_ϕ, E_r) are implemented by classical components. The input image size is $128 \times 128 \times 172$. In the image restoration task investigated in Section IV, the quantum part (i.e., the core FE module) takes the majority of total computational time (about 98.89%).

understanding, the inputs of the Ising-rotation architecture are purposely designed to come from different feature maps extracted by $f_{\boldsymbol{\psi}}$, in order to obtain more abstract quantum features in $E_{\boldsymbol{\omega}}$, as illustrated in Fig. 4. This completes the design of the core FE module $E_{\boldsymbol{\omega}}$.

Finally, although the low-rank module E_r can be implemented by the unitary operator (thus, linear operator) when simply considering those commonly used linear mixing models (LMMs) of HSI [48], [55], some HRS literature have revealed the nonlinearity of hyperspectral data for capturing some nonlinear effects (e.g., endmember variability, multiple reflections, and so on) observed in real HSI [56], [57]. In view of this, implementing E_r by classical networks would be a more natural and straightforward way to capture the nonlinearity. Therefore, instead of using single linear convolution layer to spectrally upsample $\boldsymbol{S}$ to the target image $\boldsymbol{X} = E_r(\boldsymbol{S})$ via (9), we introduce slight nonlinearity into the E_r module by using two linear convolution layers with one nonlinear activation layer (LeakyReLU) inserted in between; note that the network cannot end in a LeakyReLU layer (otherwise, the negative outputs would lead to invalid HSIs), and that the nonlinearity in real HSIs should not be too serious (otherwise, the aforementioned LMMs cannot so well approximate real HSIs [48], [55]). This can be equivalently implemented by one CB module and one pure convolution module, as detailed in Table II. The superiority of the HyperQUEEN and the necessity of the nonlinear design of the E_r module will be demonstrated by extensive experimental evidence in Section IV. During the training phase, the loss function should be designed based on the target application. As this work focuses on the image inpainting, once the target image $\boldsymbol{X}$ is obtained from E_r, it is naturally compared with the reference image $\boldsymbol{Y}$, thereby yielding the loss function $\|\boldsymbol{X} - \boldsymbol{Y}\|_F^2$ that can be minimized via the adaptive moment estimation (ADAM) optimizer [58] (whose superiority over other frequently used optimizers has been reported in [59]).

To conclude this section, experimental evidences will demonstrate the superiority of the quantum-based approach, but it is difficult to theoretically explain such a superiority due to too many intermingled factors. The exact processing stages of HyperQUEEN are detailed in Table II, which starts from the input corrupted HSI and ends in the corresponding restored HSI, where some special quantum operators (e.g., CCNOT) have been collectively detailed in Table I with exact mathematical definitions. As we believe that information embedded in different qubits should be further intermingled to generate more abstract features, HyperQUEEN applies Toffoli gate (i.e., CCNOT) on multiple qubits to entangle them together [60]. However, assuming that there do have some unknown mechanisms making the quantum module superior, the success of HyperQUEEN can be pinned down to three key factors: 1) the success of transforming the huge volume of typical hyperspectral information into a reasonable amount of qubits [see the design of (f_ψ, E_r)]; 2) the success of guaranteeing the FE so that the core quantum module E_ω can exert its value (see Theorem 2); and 3) the success of avoiding the distortion caused by the QC effect via the f_ϕ module. Theoretically understanding why HyperQUEEN works so well will be a challenging but critical future research line.

IV. Experimental Results and Discussion

In this section, we evaluate the effectiveness of the proposed HyperQUEEN architecture/theory when executing the advanced quantum image processing task, i.e., restoration of NASA's damaged hyperspectral images. Since this is the first

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quantum architecture for such an advanced task, we could just adopt classical methods as baselines, including the convex optimization-based fast hyperspectral image inpainting method (FastHyIn) [61], deep learning-based hyperspectral deep image prior (HDIP) [62], and low-rank modeling-based hyperspectral destriping method (LRHyDS) [63]. Remarkably, LRHyDS is a blind method, while the other two are nonblind. On all four representative landscapes, i.e., city, mountain, farm, and lake, the proposed HyperQUEEN shows outstanding quantitative and qualitative performances. Besides showing the necessity of incorporating the nonlinear low-rank module E_r, our experiments seem to support the superiority of the trainable Ising quantum entanglement scheme when compared with the Toffoli entanglement.

A. Experimental Settings

The baselines are briefly introduced in the following. For FastHyIn [61], it adopts a convex optimization algorithm called the alternating direction method of multiplier (ADMM) [64], for which the implicit self-similarity prior is implemented by plugging the fast block-matching 3-D (BM3D) denoiser [65] to solve the proximal mapping; the overall speed is very fast. For HDIP [62], it assumes that the deep network structure itself is a prior for regularizing the ill-posed inpainting problem, thereby judiciously reformulating the inverse problem from the hyperspectral optimization domain into the network optimization domain. For LRHyDS [63], it employs the crucial technique known as hyperspectral unmixing [2], as well as the low-rank modeling therein, to achieve hyperspectral inpainting under a blind setting without requiring the information of damaged positions ahead of time. Note that the proposed HyperQUEEN also achieves such a blind mechanism.

In our experiments, the training data collected over the North American region were acquired by NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the period from 2008 to 2018 [66]. After removing the water-vapor absorption bands (i.e., bands 1–10, 104–116, 152–170, and 215–224) from the original 224 AVIRIS bands, a total of 172 bands are used in the experiments [67]. HyperQUEEN is trained just using a small dataset, composed of just 480 training images, each containing 128 × 128 pixels. The ADAM optimizer with a learning rate of 0.01 [58] is adopted for the training stage, where the number of epochs is set as 200. The training batch size is configured as 2 because the maximum number of qubits (about 100 even in the most advanced quantum computers [21]) does not allow us to feed too much training data at once.

The training environment is constructed by NVIDIA GTX-2080Ti GPU, Intel i9-9900X CPU (3.50-GHz speed and 128-GB RAM), and Intel i9-10900X CPU (3.70-GHz speed and 64-GB RAM). We utilize the PyTorch and PennyLane (the leading tool for programming quantum computers) frameworks [51], [52] with Python 3.10.4 and Python 3.10.6 for training the HyperQUEEN. In the testing phase, the experiments are executed on a facility equipped with NVIDIA GTX-3090 GPU and AMD Ryzen 9 5950X 16-core processor (4.90-GHz speed and 128-GB RAM) with the testing environments set as MATLAB R2020b and Python 3.10.6.



Fig. 5. True color composition of a commonly investigated hyperspectral scene in Bhilwara of India.

TABLE III
COMPUTATIONAL TIME T (IN SECONDS) OF VARIOUS HYPERSPECTRAL INPAINTING METHODS UNDER TEST

Methods	Time T
HyperQUEEN	**0.037**
FastHyIn [61]	2.140
HDIP [62]	26.638
LRHyDS [63]	7.050

B. Qualitative Analysis

In this section, we test the real-hyperspectral satellite data, acquired by the Hyperion push-broom type imager aboard NASA's Earth Observation-1 (EO-1) satellite, to evaluate the effectiveness of HyperQUEEN compared to other benchmark inpainting methods. The Hyperion sensor has 242 spectral channels with wavelengths ranging from 0.4 to 2.5 μm and has a spectral resolution of approximately 10 nm. A commonly investigated region of interest (ROI) over Bhilwara of India [68] is displayed in Fig. 5. The tested real-satellite image, acquired on February 20, 2012, contains 128 × 128 pixels [69] and, after removing water-vapor bands 1–7, 61–77, 122–128, 166–178, and 217–242 [70], a total of 172 hyperspectral bands are used in our experiment. Some representative results corresponding to slight corruption (i.e., bands 99), serious corruption (i.e., band 57), and completely missing band (i.e., band 58) are displayed in Fig. 6.

One can see that for the relatively simple case of band 99 wherein the corruption pattern just contains two stripes, all the methods under test perform very well, as can be seen from the first column Fig. 6. However, for band 57 seriously corrupted by densely populated stripes in the left region, it appears to have some radiation or geometry distortions in the band images recovered by FastHyIn, HDIP, and LRHyDS, as can be seen from the second column Fig. 6. As for the most challenging case of band 58, for which the observation data are completely missing, the benchmark methods FastHyIn and LRHyDS completely fail to recover the spatial information, as can be seen from the third column of Fig. 6.

All in all, HDIP and HyperQUEEN perform best on the Hyperion satellite data. Nevertheless, by observing the computational efficiency of these methods, as shown in Table III, one

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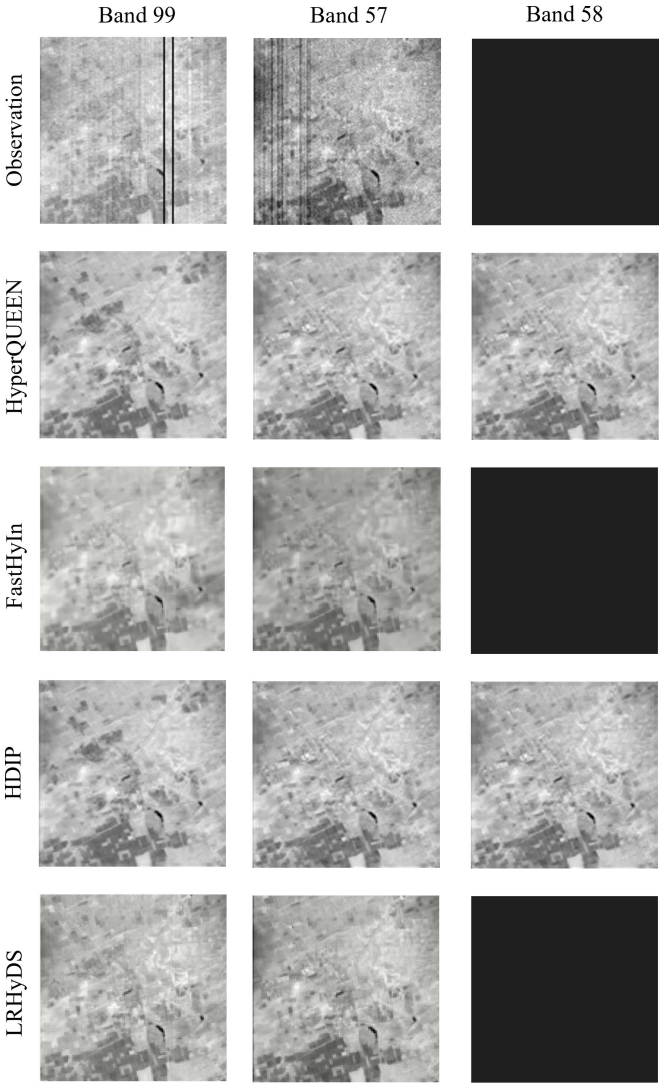


Fig. 6. Inpainting results of NASA's EO-1 Hyperion hyperspectral satellite images acquired over Bhilwara of India. The tested bands 99, 57, and 58 are some representatives corresponding to slight corruption, serious corruption, and complete corruption, respectively. The tested real-world scenarios are considered highly damaged (e.g., the completely missing band 58), for which even state-of-the-art algorithms (e.g., FastHyIn and LRHyDS) completely fail to restore the 128 × 128 image of 30-m resolution, corresponding to a large area of 3.84 × 3.84 km^2. Therefore, HyperQUEEN can well solve the challenging real-world restoration problem.

can see that the proposed HyperQUEEN is about 720 times faster than HDIP, nearly three orders of magnitude faster. This may be attributed to the nearly real-time processing of the quantum mechanisms. Due to the lack of ground truth in this experiment, we will conduct extensive quasi-real data experiments based on Wald's protocol [71], in order to fairly argue that the proposed HyperQUEEN does have superior effectiveness and computational efficiency based on well-known performance metrics in Section IV-C.

C. Quantitative Analysis

In this section, we quantitatively evaluate the performances of HyperQUEEN and other benchmark methods based on Wald's protocol [71]. To this end, we need to adopt some quantitative performance indices, including peak signal-to-noise ratio (PSNR), root-mean-square error (RMSE), spectral angle mapper (SAM), and structural similarity (SSIM) [29], [72]. Based on Wald's protocol, random dense stripes (the most common missing pattern in remote sensing images [29], [68]) are used to corrupt ground-truth images, i.e., real-hyperspectral data, and then the corrupted images are restored by inpainting methods under test, followed by computing the similarity between the restored images and the ground-truth images based on PSNR/RMSE/SAM/SSIM, thereby evaluating the effectiveness of these methods. By letting $\boldsymbol{X} \in \mathbb{R}^{M\times L}$ and $\boldsymbol{Y} \in \mathbb{R}^{M\times L}$ be the restored HSI and the reference HSI, respectively, their definitions [29], [72] are given in the following for rigorousness. First, PSNR is defined as

$$\mathrm{PSNR}(\boldsymbol{X},\boldsymbol{Y}) \triangleq \frac{1}{M}\sum_{m=1}^{M} 10\log_{10}\left(\frac{(\max\{y_{ml}|l\in\mathcal{I}_L\})^2}{\frac{1}{L}\|\boldsymbol{Y}^{(m)}-\boldsymbol{X}^{(m)}\|_F^2}\right)$$

where y_{ml} is the (m,l)th entry of $\boldsymbol{Y}$, $\mathcal{I}_L \triangleq \{1,2,\ldots,L\}$, and $\boldsymbol{X}^{(m)}, \boldsymbol{Y}^{(m)} \in \mathbb{R}^L$ denote the mth bands of $\boldsymbol{X}$ and $\boldsymbol{Y}$. Second, RMSE is defined as

$$\mathrm{RMSE}(\boldsymbol{X},\boldsymbol{Y}) \triangleq \sqrt{\frac{1}{ML}\|\boldsymbol{Y}-\boldsymbol{X}\|_F^2}.$$

Third, SAM is defined as

$$\mathrm{SAM}(\boldsymbol{X},\boldsymbol{Y}) \triangleq \frac{1}{L}\sum_{l=1}^{L}\arccos\left(\frac{[\boldsymbol{Y}]_{:,l}^T[\boldsymbol{X}]_{:,l}}{\|[\boldsymbol{Y}]_{:,l}\|\,\|[\boldsymbol{X}]_{:,l}\|}\right)$$

where $[\boldsymbol{X}]_{:,l}, [\boldsymbol{Y}]_{:,l} \in \mathbb{R}^M$ denote the lth pixels of $\boldsymbol{X}$ and $\boldsymbol{Y}$. Fourth, SSIM is defined as $\mathrm{SSIM}(\boldsymbol{X},\boldsymbol{Y}) \triangleq (1/M)\sum_{m=1}^{M}\mathrm{SSIM}_m$, where

$$\mathrm{SSIM}_m \triangleq \mathbb{E}\left[\frac{2\mu_x\mu_y+C_1}{\mu_x^2+\mu_y^2+C_1}\cdot\frac{2\sigma_x\sigma_y+C_2}{\sigma_x^2+\sigma_y^2+C_2}\cdot\frac{\sigma_{xy}^2+C_3}{\sigma_x\sigma_y+C_3}\right]$$

in which $C_3 = C_2/2$ with all the C_i's being small nonnegative constants for avoiding unstable results; $\boldsymbol{y}$ is the mth reference band $\boldsymbol{Y}^{(m)}$ restricted in a 7 × 7 sliding window, and $\boldsymbol{x}$ is defined similarly; $\mathbb{E}$ is an averaging operator over windows; μ_x and μ_y are the means of $\boldsymbol{x}$ and $\boldsymbol{y}$, respectively; σ_x and σ_y are the standard deviations of $\boldsymbol{x}$ and $\boldsymbol{y}$, respectively; and σ_{xy}^2 is the covariance between $\boldsymbol{x}$ and $\boldsymbol{y}$. Though the definition looks complicated, the three terms in SSIM have clear physical meanings, which correspond to luminance distortion, contrast distortion, and loss of correlation, respectively. Higher values of PSNR and SSIM indicate better performances, while lower values of RMSE and SAM indicate better results.

The real-hyperspectral data here are NASA's AVIRIS data introduced in Section IV-A, and we test four representative landscapes (i.e., city, mountain, farm, and lake) with the 128 × 128 ROIs shown in the four top-left corners of Figs. 7–10, where the RGB bands are band 29, band 21, and band 11, respectively. As for the above-mentioned stripe patterns, we randomly remove about 10% of the total 128 lines at each spectral band, and some samples of the resultant stripe missing patterns are displayed in the second columns of Figs. 7–10. Accordingly, the quantitative analysis results are summarized in Tables IV–VII, where boldfaced numbers

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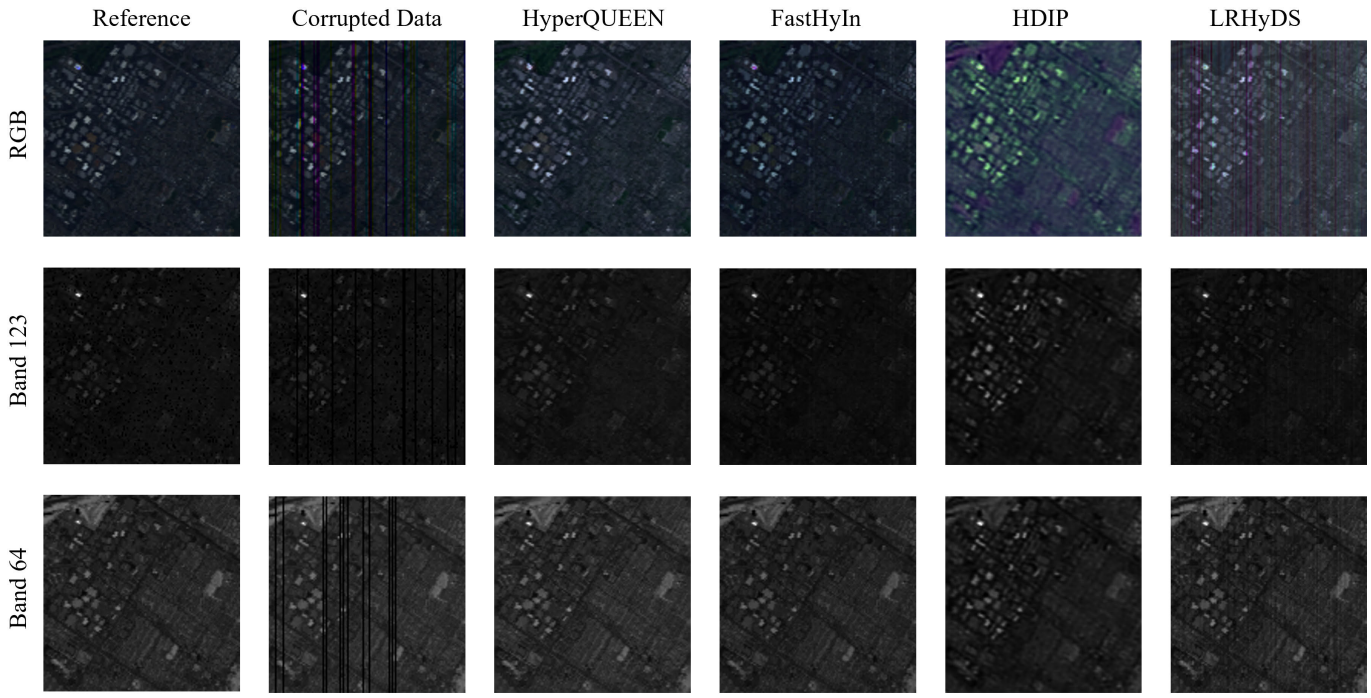


Fig. 7. Inpainting results of some representative bands, i.e., RGB bands, band 123 with uniform stripes, and band 64 with dense stripes, for the hyperspectral data acquired over the city landscape.

TABLE IV
QUANTITATIVE ANALYSIS OF HYPERQUEEN AND THREE BENCHMARK METHODS ON CITY LANDSCAPE DATA

Methods	PSNR (↑)	RMSE (↓)	SAM (↓)	SSIM (↑)	Time T
HyperQUEEN	**38.918**	**0.008**	**3.557**	**0.979**	**0.026**
FastHyIn [61]	34.613	0.021	11.724	0.976	1.800
HDIP [62]	28.998	0.021	8.765	0.936	25.760
LRHyDS [63]	32.593	0.015	6.439	0.954	5.820

TABLE V
QUANTITATIVE ANALYSIS OF HYPERQUEEN AND THREE BENCHMARK METHODS ON MOUNTAIN LANDSCAPE DATA

Methods	PSNR (↑)	RMSE (↓)	SAM (↓)	SSIM (↑)	Time T
HyperQUEEN	**32.984**	**0.016**	**1.731**	0.965	**0.026**
FastHyIn [61]	27.145	0.040	5.382	0.909	1.580
HDIP [62]	29.931	0.018	2.448	**0.969**	22.060
LRHyDS [63]	26.561	0.037	5.842	0.911	5.350

TABLE VI
QUANTITATIVE ANALYSIS OF HYPERQUEEN AND THREE BENCHMARK METHODS ON FARM LANDSCAPE DATA

Methods	PSNR (↑)	RMSE (↓)	SAM (↓)	SSIM (↑)	Time T
HyperQUEEN	**31.014**	**0.017**	**3.538**	**0.967**	**0.025**
FastHyIn [61]	18.067	0.075	16.042	0.713	1.670
HDIP [62]	28.568	0.022	4.631	0.950	22.115
LRHyDS [63]	24.759	0.031	5.506	0.939	5.540

TABLE VII
QUANTITATIVE ANALYSIS OF HYPERQUEEN AND THREE BENCHMARK METHODS ON LAKE LANDSCAPE DATA

Methods	PSNR (↑)	RMSE (↓)	SAM (↓)	SSIM (↑)	Time T
HyperQUEEN	**30.794**	0.014	**3.468**	**0.985**	**0.025**
FastHyIn [61]	30.272	**0.013**	5.357	0.971	1.570
HDIP [62]	23.028	0.021	6.422	0.915	21.971
LRHyDS [63]	24.407	0.027	9.972	0.939	5.610

indicate the best results. One can see that for the city and farm landscapes, the proposed HyperQUEEN performs best in terms of all the four indices, as can be seen from Tables IV and VI. For the mountain data, Table V shows that HyperQUEEN also performs best in terms of PSNR/RMSE/SAM and nearly the best in terms of SSIM. For the lake data, Table VII shows that HyperQUEEN yields the best performance in terms of PSNR/RMSE/SSIM and nearly the best in terms of SAM. The truly outstanding quantitative analysis results double confirm the high effectiveness of the design philosophies and principles behind the proposed HyperQUEEN.

We should remark that HDIP has the best performance in terms of SSIM for the mountain data (see. Table V), probably because the deep image prior embedded in the deep network is essential to capture the complicated mountainous texture. The deep learning method HDIP does achieve good performance for the investigated challenging hyperspectral scenes of serious stripes, as it clearly outperforms other classical methods in the mountain and farm data (see Tables V and VI). Even though, the introduced quantum mechanisms in HyperQUEEN still show clear superiority over HDIP for all the landscapes, with a significant lead in PSNR and with much faster speed. On the other hand, FastHyIn has outstanding spectral reconstruction ability, as evidenced by the lowest SAM value for the lake data (see Table VII), and this may be attributed to the unmixing modeling adopted therein. From the results of

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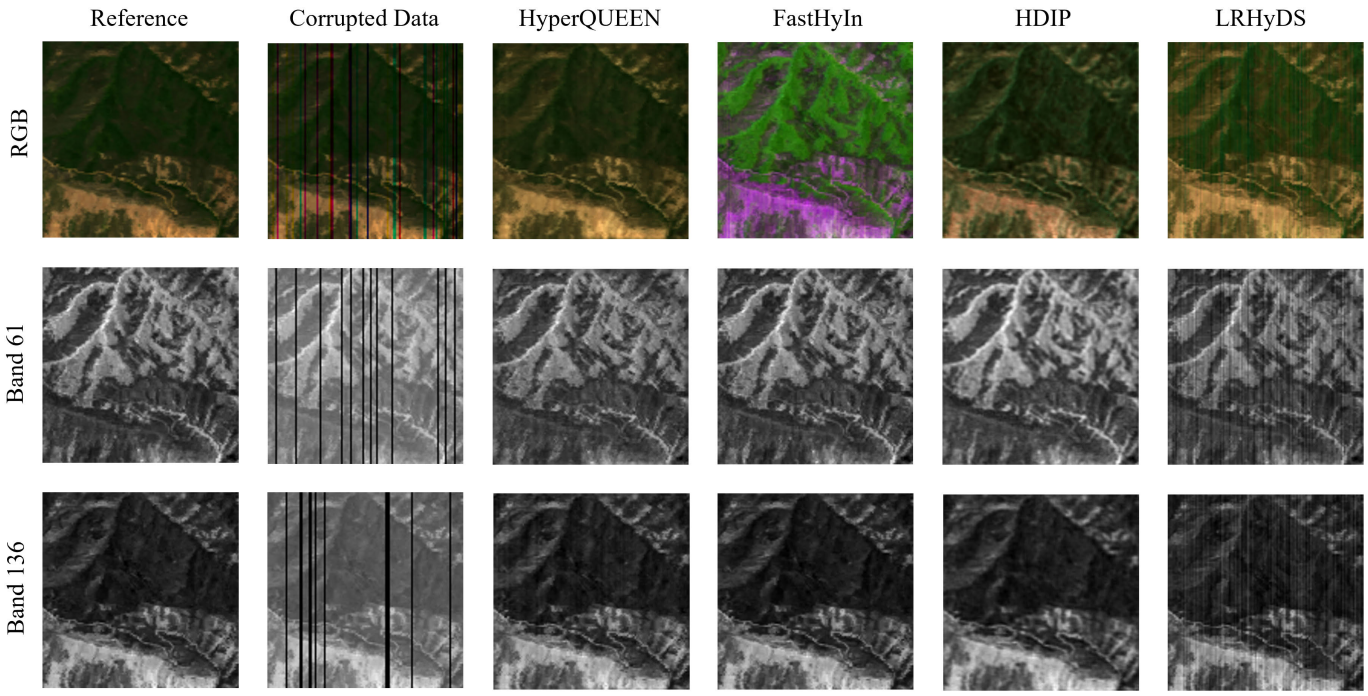


Fig. 8. Inpainting results of some representative bands, i.e., RGB bands, band 61 with uniform stripes, and band 136 with dense stripes, for the hyperspectral data acquired over the mountain landscape.

the blind method, i.e., LRHyDS, one can see that achieving the advanced image processing task (hyperspectral inpainting) under a blind setting is rather challenging. Even so, the blind mechanism of HyperQUEEN still exhibits its superiority in terms of all four performance indices.

Although this section is mainly an endeavor at quantitative analysis, the visual reconstruction results of HyperQUEEN and the other three methods are also provided for better understanding, as can be seen from the third to sixth columns of Figs. 7–10. For city data, only HyperQUEEN and FastHyIn have successfully restored the image, while HDIP has obvious color distortion, as can be seen from Fig. 7, from which one can also see that this scenario seems too challenging for the blind method LRHyDS. In Fig. 8, FastHyIn exhibits serious color distortion in the reconstructed result, probably because it is too difficult to search many similar patches for BM3D regularization for such a highly complicated/aperiodic structure of mountainous texture. Again this complicated texture is also too challenging for the blind method, i.e., LRHyDS, as can be seen from the last column of Fig. 8. For Fig. 9, the texture is also relatively complicated, so the reconstruction results of all four tested methods are not perfect. However, the proposed HyperQUEEN has the best visual quality for the farm data; see the true color compositions in the last row of Fig. 9. Finally, from Fig. 10, HyperQUEEN still yields the best visual results, while all the other three methods suffer from serious color distortions.

Therefore, the proposed HyperQUEEN has the best visual and quantitative analysis results in general, besides its outstanding computational efficiency as reported in Tables IV–VII, where the parallel computational time (w.r.t. the GPU/CPU specified in Section IV-A) is measured in seconds. Among the three peer methods, FastHyIn is truly fast, thanks to the fast BM3D regularization scheme adopted therein. However, quantum computing-based HyperQUEEN is still pretty much faster than FastHyIn since the involved quantum groups can be parallelly realized with an in-group quantum entanglement mechanism. Specifically, averaging over the four landscapes, HyperQUEEN is about two orders of magnitude faster than FastHyIn.

D. In-Depth Study of Low-Rank Module and Quantum Entanglement Mechanisms

HyperQUEEN involves the DC module f_{ψ}, the core FE module E_{ω}, the inverse-QC module f_{ϕ}, and the low-rank E_r module. The DC module is needed as it allows quantum processing to be executed using the very limited qubits in current quantum computers, while the inverse-QC module is also needed as the QC effect does not allow us to directly retrieve the QSTI. As for the other two modules, however, we would like to know the effectiveness of the E_r module and to understand more about the entanglement mechanisms that happened in the core FE module E_{ω}.

As discussed in Section III, there are some nonlinear effects in real-hyperspectral data [56], [57], so, to reconstruct high-fidelity hyperspectral images, we need to introduce nonlinear mechanisms into the HyperQUEEN architecture. For this reason, we purposely insert a nonlinear activation layer (i.e., LeakyReLU) into two linear convolution layers, and this version is denoted by "HyperQUEEN(E_r^1)" meaning that the E_r module is on. To understand its effectiveness, another version denoted by "HyperQUEEN(E_r^0)" turns off the E_r module; specifically, the nonlinear layer is removed, and the two linear layers are retained (and equivalently merged into a single-linear layer) to transform the collapsed quantum features

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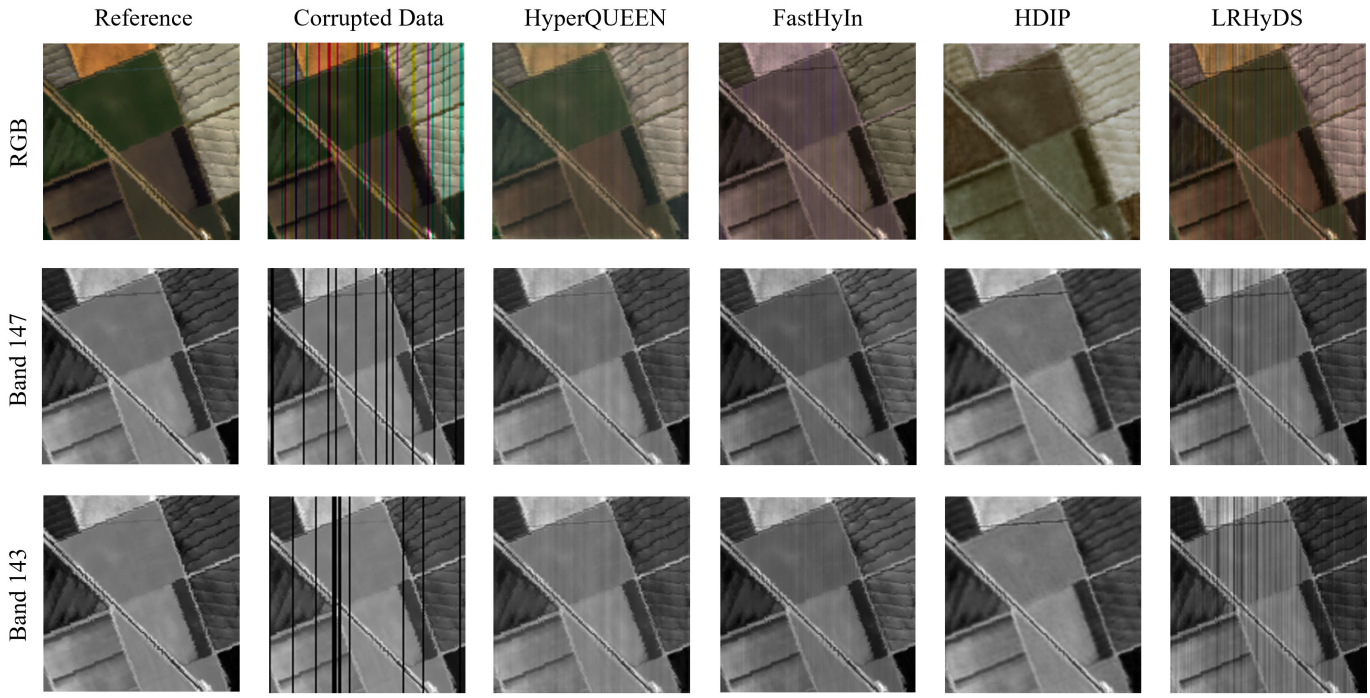


Fig. 9. Inpainting results of some representative bands, i.e., RGB bands, band 147 with uniform stripes, and band 143 with dense stripes, for the hyperspectral data acquired over farm landscape.

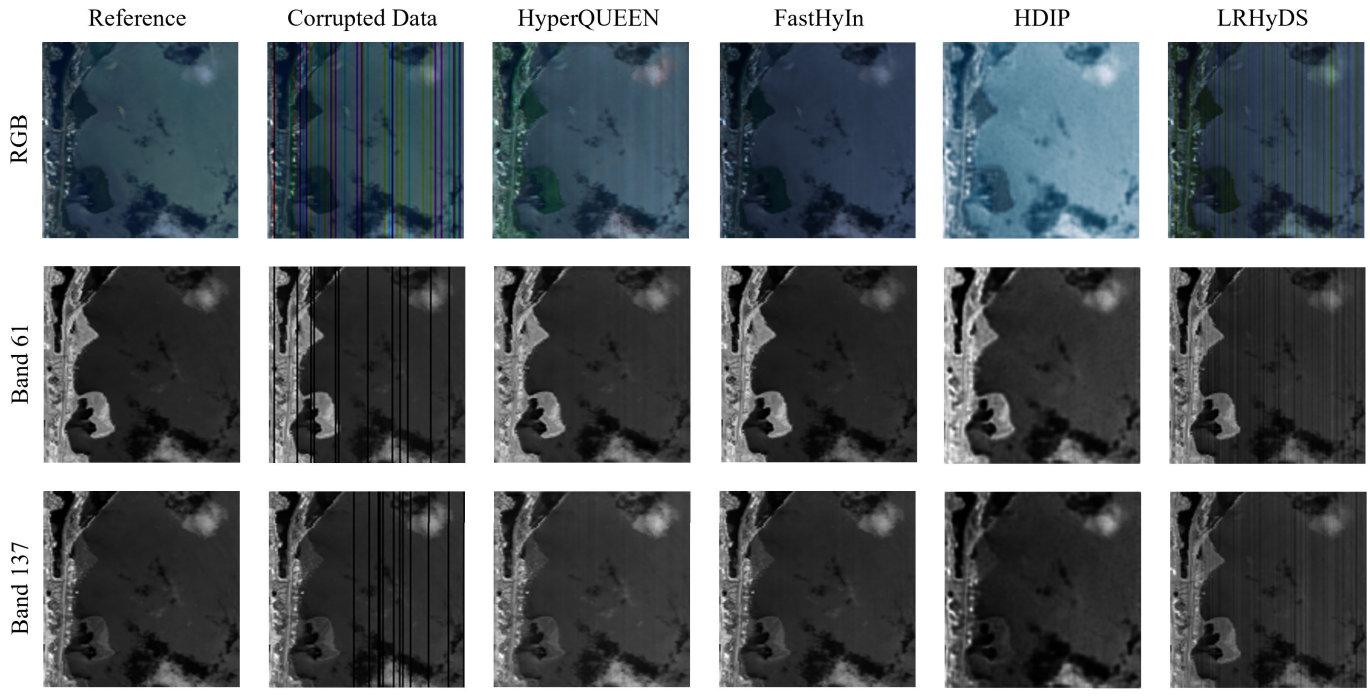


Fig. 10. Inpainting results of some representative bands, i.e., RGB bands, band 61 with uniform stripes, and band 137 with dense stripes, for the hyperspectral data acquired over lake landscape.

back to the original hyperspectral domain. Their comparisons averaged over all the testing data across four landscapes are summarized in Table VIII. One can see that besides the SAM index, for which both versions have equivalently good spectral reconstruction qualities, the version of HyperQUEEN(E_r^1) does yield superior performance in terms of PSNR, RMSE, and SSIM, showing the necessity of including the low-rank E_r module in HyperQUEEN.

Finally, it is known that the quantum entanglement mechanism is essential to quantum computing, but our experiments in the following further reveal that over-entanglement does not seem to be a good strategy. Recall that there are

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TABLE VIII
EVALUATION OF THE EFFECTIVENESS OF THE LOW-RANK MODULE E_r

Modes	PSNR (↑)	RMSE (↓)	SAM (↓)	SSIM (↑)
HyperQUEEN(E_r^1)	**35.597**	**0.013**	**2.493**	**0.973**
HyperQUEEN(E_r^0)	34.746	**0.013**	2.584	0.970

TABLE IX
COMPARISONS AMONG DIFFERENT ENTANGLEMENT MECHANISMS IN THE CORE FE MODULE E_ω

Modes	PSNR (↑)	RMSE (↓)	SAM (↓)	SSIM (↑)
HyperQUEEN(I_1, T_1)	35.597	0.013	2.493	0.973
HyperQUEEN(I_0, T_1)	**36.158**	**0.012**	2.434	**0.977**
HyperQUEEN(I_1, T_0)	35.776	**0.012**	**2.365**	**0.977**
HyperQUEEN(I_0, T_0)	34.363	0.014	2.660	0.970

Ising entanglement [i.e., $XX(\theta)$] and Toffoli entanglement (i.e., CCNOT) in the proposed HyperQUEEN architecture, as depicted in Fig. 4. Thus, there are four combinations "HyperQUEEN(I_1, T_1)," "HyperQUEEN(I_0, T_1)," "HyperQUEEN(I_1, T_0)," and "HyperQUEEN(I_0, T_0)," where I_1 (respectively, I_0) means that Ising entanglement is turned on (respectively, off), and T_1 (respectively, T_0) means that Toffoli entanglement is turned on (respectively, off). For example, HyperQUEEN(I_1, T_0) denotes the Ising-on Toffoli-off mode. The experimental results are presented in Table IX.

First, we focus on the Ising entanglement by turning off the Toffoli entanglement and find that HyperQUEEN(I_1, T_0) is always better than HyperQUEEN(I_0, T_0) in terms of all the four performance indices, confirming that Ising entanglement has exerted its role in quantum processing. Second, we focus on the Toffoli entanglement by turning off the Ising entanglement, leading to the comparison between the two modes HyperQUEEN(I_0, T_1) and HyperQUEEN(I_0, T_0). One can observe that the mode HyperQUEEN(I_0, T_1) always yields better performances than the mode HyperQUEEN(I_0, T_0) in terms of all the four performance indices, showing that Toffoli entanglement is also useful.

The above-mentioned two observations may cause a misconception, which we call "over-entanglement;" this term, not seen in the existing literature, is invented by ourselves to reflect our empirical experience that more entanglement mechanisms would not imply better results (even if all the used entanglement mechanisms are already proved useful). Specifically, based on the above-mentioned two observations (i.e., both Ising and Toffoli entanglement mechanisms are useful), one may intuitively guess that turning on both Ising and Toffoli entanglement mechanisms [i.e., the HyperQUEEN(I_1, T_1) mode] can further upgrade the performance because merely turning on one of them [i.e., HyperQUEEN(I_0, T_1) or HyperQUEEN(I_1, T_0)] does lead to better results (as discussed earlier). Nevertheless, for this advanced quantum image processing task, we somewhat surprisingly find that HyperQUEEN(I_1, T_1) actually yields weaker results than both modes of HyperQUEEN(I_0, T_1) and HyperQUEEN(I_1, T_0). Although this is quite counterintuitive, our empirical observation suggests future researchers working on quantum deep remote sensing to avoid over-entanglement in the architecture design.

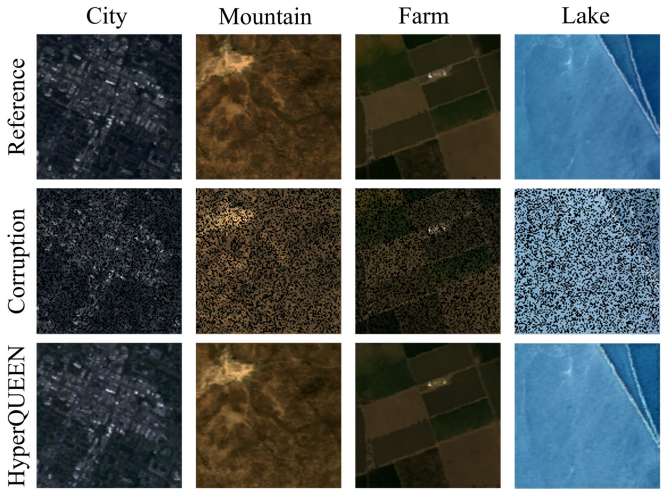


Fig. 11. Qualitative analysis for the application of HyperQUEEN to HSI pepper noise removal.

TABLE X
QUANTITATIVE ANALYSIS FOR THE APPLICATION OF HYPERQUEEN TO HSI PEPPER NOISE REMOVAL

Landscapes	PSNR (↑)	RMSE (↓)	SAM (↓)	SSIM (↑)	Time T
City	34.462	0.013	3.163	0.963	0.026
Mountain	34.681	0.013	1.265	0.946	0.024
Farm	35.860	0.004	1.807	0.973	0.022
Lake	34.712	0.008	1.079	0.984	0.023

E. Application of HyperQUEEN to Hyperspectral Pepper Noise Removal

In this experiment, we use exactly the same HyperQUEEN architecture to solve another inverse problem, known as pepper noise removal for HSI. The experimental settings are the same as those described in Section IV-A, including the 480 training images, the loss function, the training environment, and computer, as well as the setting of the ADAM optimizer. The only difference is that the HSIs are now corrupted by serious pepper noise [73]. Specifically, there are 40% randomly selected pixels corrupted by pepper noise, as depicted in the second row of Fig. 11, where the first row shows the corresponding reference HSIs. After HyperQUEEN learns the mapping between the corrupted HSIs to the clean reference HSIs via ADAM optimizer, it is applied to restore the damaged HSIs over four representative landscapes. The visual results are shown in the third row of Fig. 11, while the quantitative evaluations of the restoration performance are summarized in Table X. One can see that the reconstructed HSIs exhibit high-fidelity visual reconstruction, as double confirmed by the good quantitative performances in terms of both reconstruction accuracy and computational efficiency. This experiment well illustrates the generalizability of HyperQUEEN, which is originally designed for image inpainting but works nicely for the pepper noise removal task. Thus, HyperQUEEN does create a new door for solving challenging inverse problems in the upcoming quantum space era.

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V. Conclusion

Quantum science winning the 2022 Nobel Prize in Physics has gained considerable attention in all computing areas and is expected to lead future development of remote sensing technologies. However, due to some practical limitations in current quantum computers (e.g., a limited number of qubits, and QC effect), existing quantum image processing methods are far from being applicable to advanced tasks, such as restoring damaged hyperspectral images of large data volume in satellite remote sensing. The proposed HyperQUEEN architecture/theory further introduces deep learning into quantum computing to solve the aforementioned practical limitations, besides employing the low-rank nature of hyperspectral data geometry. As revealed by Theorem 1, the quantum BPs effect (analogous to the GV effect in classical deep learning) suggests that avoiding too many layers and/or adding some skip connections would be good strategies for the implementation of the HyperQUEEN architecture, where over-entanglement should also be avoided (see Section IV-D). HyperQUEEN highly compresses the input image into very low-dimensional qubit states, which are then processed using the proposed five-layer core Ising-rotation quantum module with provable FE of all quantum operators; see Theorem 2. Though the processed states will collapse to some eigenstates once they are measured, HyperQUEEN is able to infer the target image from the retrieved Pauli-collapsed quantum statistics. Implementation of HyperQUEEN on real quantum computers requires cross-disciplinary collaborators with quantum physics backgrounds to work with us to achieve this exciting and impactful future research line.

To show the merit of our design of HyperQUEEN, it is applied to solve the challenging blind tensor completion problem without using the information of damaged positions, for blindly achieving quantum hyperspectral image inpainting on NASA's real data. This means a lot for the upcoming quantum space era. Though we are not aiming at developing state-of-the-art methods due to the immature quantum facility nowadays, HyperQUEEN exhibits its superior performance even when compared with those nonblind methods, well demonstrating the feasibility of quantum HRS. Since this is the first work making a breakthrough to successfully achieve advanced/complicated quantum image processing tasks, there is no reference for our design, meaning that the proposed HyperQUEEN is designed from scratch. The proposed HyperQUEEN architecture and the reported design philosophies/principles serve as an informative guide for future researchers to work on quantum deep learning and quantum remote sensing.

Appendix

A. Proof of Theorem 1

By (10), we can express the gradient explicitly as

$$\frac{\partial J}{\partial \omega_i} = \frac{\partial \tilde{L}\begin{pmatrix} E_1(f_{\boldsymbol{\psi}}(x),\boldsymbol{\omega}) \\ E_2(f_{\boldsymbol{\psi}}(x),\boldsymbol{\omega}) \\ \vdots \\ E_k(f_{\boldsymbol{\psi}}(x),\boldsymbol{\omega}) \end{pmatrix}}{\partial \omega_i} \tag{11}$$

where $\tilde{L} \triangleq E_r \circ f_{\boldsymbol{\phi}}$ and $J \triangleq \tilde{L} \circ E_{\boldsymbol{\omega}} \circ f_{\boldsymbol{\psi}}$ are defined just for notational simplicity. By Definition 1, it suffices to prove $\boldsymbol{E}(\partial J/\partial \omega_i) \to 0$ and $\boldsymbol{Var}(\partial J/\partial \omega_i) \to 0$ as $n \to \infty$, which will be built in Step 2 and Step 3, respectively.

Step 1: By (11) and (7), and the assumption of t-design with $t = 1$, we have

$$\boldsymbol{E}\left(\frac{\partial J}{\partial \omega_i}\right) = \boldsymbol{E}\left(\sum_{l=1}^{k} \frac{\partial \tilde{L}}{\partial E_l}\frac{\partial E_l}{\partial \omega_i}\right) = \sum_{l=1}^{k} \boldsymbol{E}(X_l) \tag{12}$$

$$\boldsymbol{E}\left(\frac{\partial E_l(\boldsymbol{\omega})}{\partial \omega_i}\right) = 0 \tag{13}$$

where $X_l := (\partial \tilde{L}/\partial E_l)(\partial E_l/\partial \omega_i)$ in (12). Also, the assumption of independent measuring probes means that $\{X_l\}_{l=1}^{k}$ are independent random variables, that is,

$$X_{l_1} \perp\!\!\!\perp X_{l_1}, \quad \text{for } l_1 \neq l_2. \tag{14}$$

Then, by the premise of the Lipschitz decoding scheme and (13), we have $|\boldsymbol{E}(\partial \tilde{L}/\partial E_l)\boldsymbol{E}((\partial E_l(\boldsymbol{\omega}))/\partial \omega_i)| \leq \alpha 0 = 0$, implying that $\boldsymbol{E}((\partial \tilde{L}/\partial E_l)(\partial E_l/\partial \omega_i)) = \boldsymbol{Cov}((\partial \tilde{L}/\partial E_l), (\partial E_l/\partial \omega_i)) + \boldsymbol{E}(\partial \tilde{L}/\partial E_l)\boldsymbol{E}(\partial E_l/\partial \omega_i) = \boldsymbol{Cov}((\partial \tilde{L}/\partial E_l), (\partial E_l/\partial \omega_i))$. So, from (12), we conclude that

$$\boldsymbol{E}\left(\frac{\partial J}{\partial \omega_i}\right) = \sum_{l=1}^{k} \boldsymbol{Cov}\left(\frac{\partial \tilde{L}}{\partial E_l}, \frac{\partial E_l}{\partial \omega_i}\right) \tag{15}$$

which approaches 0 as will be seen next.

Step 2: By the assumption of Lipschitz decoder (i.e., $\|\nabla_{E_{\boldsymbol{\omega}}}(E_r \circ f_{\boldsymbol{\phi}})\| = \|\nabla_{E_{\boldsymbol{\omega}}}\tilde{L}\| \leq \alpha$), we know that the quantity

$$\left|\frac{\partial \tilde{L}}{\partial E_l}\right| \leq \alpha \tag{16}$$

is also bounded. As a well-known result in probability theory, for a random variable in the range of $[a, b]$, its variance can be proved to be bounded by $(b-a)^2/4$; therefore, (16) just implies that $\boldsymbol{Var}(\partial \tilde{L}/\partial E_l) \leq [\alpha - (-\alpha)]^2/4 = \alpha^2$. Moreover, using (8) and the assumption of t-design with $t = 2$, we see

$$\boldsymbol{Var}\left(\frac{\partial E_l(\boldsymbol{\omega})}{\partial \omega_i}\right) \to 0. \tag{17}$$

By employing another well-known inequality in probability theory, i.e., $|\boldsymbol{Cov}(X, Y)| \leq \sqrt{\boldsymbol{Var}(X)\boldsymbol{Var}(Y)}$, we have

$$\left|\boldsymbol{Cov}\left(\frac{\partial \tilde{L}}{\partial E_l}, \frac{\partial E_l}{\partial \omega_i}\right)\right| \leq \sqrt{\alpha^2 \boldsymbol{Var}\left(\frac{\partial E_l(\boldsymbol{\omega})}{\partial \omega_i}\right)} \tag{18}$$

which approaches to $\alpha\sqrt{0} = 0$ [see (17)]. Therefore, having (15) in mind, the proof of the first desired result, i.e., $\boldsymbol{E}(\partial J/\partial \omega_i) \to 0$, just follows from (18).

Step 3: The remaining task is to prove that $\boldsymbol{Var}(\partial J/\partial \omega_i) \to 0$. To this end, we first prove that $\boldsymbol{Var}((\partial \tilde{L}/\partial E_l)(\partial E_l/\partial \omega_i)) \to 0$. By (16), we have

$$\boldsymbol{Var}\left(\frac{\partial \tilde{L}}{\partial E_l}\frac{\partial E_l}{\partial \omega_i}\right) \leq \boldsymbol{E}\left(\left(\frac{\partial \tilde{L}}{\partial E_l}\right)^2\left(\frac{\partial E_l}{\partial \omega_i}\right)^2\right) \leq \alpha^2 \boldsymbol{E}\left(\left(\frac{\partial E_l}{\partial \omega_i}\right)^2\right). \tag{19}$$

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Then, by jointly using (13) and (17), we have $\boldsymbol{E}((\partial E_l/\partial\omega_i)^2) = \boldsymbol{Var}(\partial E_l/\partial\omega_i) + \boldsymbol{E}(\partial E_l/\partial\omega_i)^2 = \boldsymbol{Var}(\partial E_l/\partial\omega_i) \to 0$, which, together with (19), yields the following result, that is,

$$\boldsymbol{Var}\left(\frac{\partial\tilde{L}}{\partial E_l}\frac{\partial E_l}{\partial\omega_i}\right) \to 0. \tag{20}$$

By further employing (11), we obtain

$$\boldsymbol{Var}\left(\frac{\partial J}{\partial\omega_i}\right) = \boldsymbol{Var}\left(\sum_{l=1}^{k}\frac{\partial\tilde{L}}{\partial E_l}\frac{\partial E_l}{\partial\omega_i}\right).$$

For independent random variables, the variance of their sum is the sum of their variances, so (14) implies that

$$\boldsymbol{Var}\left(\frac{\partial J}{\partial\omega_i}\right) = \sum_{l=1}^{k}\boldsymbol{Var}\left(\frac{\partial\tilde{L}}{\partial E_l}\frac{\partial E_l}{\partial\omega_i}\right)$$

which together with (20), yields the second desired result, i.e., $\boldsymbol{Var}(\partial J/\partial\omega_i) \to \sum_{l=1}^{k} 0 = 0$. Therefore, the proof of Theorem 1 has been completed. ■

B. Proof of Theorem 2

We first build two lemmas, one for the transforms among quantum rotation gates and the other for unitary matrix decomposition, followed by using them to prove Theorem 2.

Step 1: By defining the auxiliary complex-valued matrix $\boldsymbol{C} = (1/2)\begin{pmatrix}1-i & -1-i\\ 1-i & 1+i\end{pmatrix}$, some useful transforms are summarized in the following lemma.

Lemma 1: For any real-valued θ, we have

$$\begin{aligned} R_Y(\theta) &= \boldsymbol{C}R_X(\theta)\boldsymbol{C}^\dagger\\ R_Z(\theta) &= \boldsymbol{C}R_Y(\theta)\boldsymbol{C}^\dagger\\ R_X(\theta) &= \boldsymbol{C}R_Z(\theta)\boldsymbol{C}^\dagger \end{aligned}$$

where the quantum rotation gates are defined in Table I. □

Proof: To show the first equality, recall the Pauli-Y gate $Y = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}$ and the Pauli-X gate $X = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$. Then, we have the well-known relation between the two quantum gates, i.e., $Y = \boldsymbol{C}X\boldsymbol{C}^\dagger$ [44, Sec. 3.6], implying that

$$\begin{aligned} \boldsymbol{C}R_X(\theta)\boldsymbol{C}^\dagger &= \boldsymbol{C}\left(\cos\frac{\theta}{2}\boldsymbol{I} - i\sin\frac{\theta}{2}X\right)\boldsymbol{C}^\dagger\\ &= \cos\frac{\theta}{2}\boldsymbol{I} - i\sin\frac{\theta}{2}Y\\ &= R_Y(\theta) \end{aligned}$$

thereby completing the proof of the first transform. The proof for the other two transforms can be done through a similar procedure and is omitted for conciseness. ■

Step 2: The following unitary matrix decomposition is useful in the main proof.

Lemma 2: If W is a unitary operator on a single qubit, then there exist real-valued numbers α, β, γ, and δ, such that

$$W = e^{i\alpha}R_Y(\beta)R_X(\gamma)R_Y(\delta)$$

where the quantum rotation gates are defined in Table I. □

Proof: Define the unitary matrix $V = \boldsymbol{C}W\boldsymbol{C}^\dagger$. By [41, Th. 4.1], for any unitary matrix V, there exist real-valued numbers α, β, γ, and δ, such that

$$V = e^{i\alpha}R_Z(\beta)R_Y(\gamma)R_Z(\delta).$$

Then, by Lemma 1, we have

$$\begin{aligned} V &= e^{i\alpha}(\boldsymbol{C}R_Y(\beta)\boldsymbol{C}^\dagger)(\boldsymbol{C}R_X(\gamma)\boldsymbol{C}^\dagger)(\boldsymbol{C}R_Y(\delta)\boldsymbol{C}^\dagger)\\ &= \boldsymbol{C}\left(e^{i\alpha}R_Y(\beta)R_X(\gamma)R_Y(\delta)\right)\boldsymbol{C}^\dagger \end{aligned}$$

implying that $W = \boldsymbol{C}^\dagger V\boldsymbol{C} = e^{i\alpha}R_Y(\beta)R_X(\gamma)R_Y(\delta)$. ■

Step 3: By the definition of the Ising XX gate $XX(\theta)$ in Table I, as well as the definition of $XX_{mn}(\theta)$ in Theorem 2, the setting of the real-valued $\theta_k := 0$ yields

$$\begin{aligned} I_0(\theta_0,\theta_1) &= XX_{03}(0)\otimes XX_{12}(0) = (\boldsymbol{I}_2\otimes\boldsymbol{I}_2)\otimes(\boldsymbol{I}_2\otimes\boldsymbol{I}_2)\\ I_1(\theta_2,\theta_3) &= XX_{01}(0)\otimes XX_{23}(0) = (\boldsymbol{I}_2\otimes\boldsymbol{I}_2)\otimes(\boldsymbol{I}_2\otimes\boldsymbol{I}_2) \end{aligned}$$

implying that we just need to find real-valued $(\alpha_k,\beta_k,\gamma_k,\delta_k)$ such that $U = ABC$. However, this is a direct result of Lemma 2, which ensures the existence of $(\alpha_k,\beta_k,\gamma_k,\delta_k)$ satisfying

$$U_k = e^{i\alpha_k}R_Y(\beta_k)R_X(\gamma_k)R_Y(\delta_k), \quad k = 0,\ldots,3.$$

Then, we have

$$U = \bigotimes_{k=0}^{3}\left[e^{i\alpha_k}R_Y(\beta_k)R_X(\gamma_k)R_Y(\delta_k)\right] = \mathrm{ABC}$$

where the last equality follows from the definitions of (A,B,C) and some basic property of Kronecker product [i.e., $(\boldsymbol{M}_1\otimes\boldsymbol{M}_2)(\boldsymbol{M}_3\otimes\boldsymbol{M}_4) = (\boldsymbol{M}_1\boldsymbol{M}_3)\otimes(\boldsymbol{M}_2\boldsymbol{M}_4)$ for matrices $\boldsymbol{M}_1,\ldots,\boldsymbol{M}_4$]. Therefore, Theorem 2 has been proven. ■

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Chia-Hsiang Lin (Member, IEEE) received the B.S. degree in electrical engineering and the Ph.D. degree in communications engineering from National Tsing Hua University (NTHU), Hsinchu, Taiwan, in 2010 and 2016, respectively.

From 2015 to 2016, he was a Visiting Student with Virginia Tech, Arlington, VA, USA. He held research positions with The Chinese University of Hong Kong, Hong Kong, from 2014 to 2017, NTHU from 2016 to 2017, and the University of Lisbon (ULisboa), Lisbon, Portugal from 2017 to 2018. He was an Assistant Professor with the Center for Space and Remote Sensing Research, National Central University, Taoyuan, Taiwan, in 2018, and a Visiting Professor with ULisboa, in 2019. He is currently an Associate Professor with the Department of Electrical Engineering, National Cheng Kung University (NCKU), Tainan, Taiwan, where he is also with the Miin Wu School of Computing. He received the Ministry of Science and Technology Young Scholar Fellowship for 2018–2023. His research interests include network science, quantum computing, convex geometry and optimization, blind signal processing, and imaging science.

Dr. Lin was a recipient of the Outstanding Doctoral Dissertation Award from the Chinese Image Processing and Pattern Recognition Society and the Best Doctoral Dissertation Award from the IEEE GRS-S in 2016. He received the EINSTEIN Grant Award in 2018. He received the Third Place from the Advances in Image Manipulation (AIM) Real World Super-Resolution Challenge at the IEEE International Conference on Computer Vision, in 2019, the Top Performance Award from the Social Media Prediction Challenge at the *Association for Computing Machinery* (ACM) *Multimedia*, in 2020, the Prize Paper Award from IEEE Geoscience and Remote Sensing Society (GRS-S), in 2020, the Best Young Professional Member Award from the IEEE Tainan Section, in 2021, the Outstanding Youth Electrical Engineer Award from The Chinese Institute of Electrical Engineering, in 2022, and the Future Technology Award from the National Science and Technology Council, in 2022.



You-Yao Chen (Student Member, IEEE) received the B.S. degree from the Department of Electrical Engineering, National Cheng Kung University, Tainan, Taiwan, in 2022, where he is currently pursuing the Ph.D. degree with the Intelligent Hyperspectral Computing Laboratory, Institute of Computer and Communication Engineering.

His research interests include convex optimization, deep learning, quantum machine learning, biomedical data analysis, and hyperspectral remote sensing.

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